

SUPPLEMENT TO “NEURAL BELLMAN OPERATORS”

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This supplement records the complete prospective-service registry, numerical semantics, confidence intervals, datewise error accounts and source-preservation map for Neural Bellman Operators. The active mathematical statements are proved in the article; all preceding proofs and applications remain in the complete companions.

1. EXECUTION, NUMERICAL SEMANTICS AND COMPLETE RECORDS

The active article retains the constructive own-future NBO operator and adds trained neural action search, non-tensor verification, prospective economic stopping and validation-qualified learned structure. Its mathematical statements are proved in the article. This supplement records the execution contract and all service outcomes. The complete development and proof companions retain every preceding theorem, application and adverse experiment under its original assumptions.

1.1. *Source and outcome identity*

The controlling advisory report reviews manuscript commit 00e837adb431f4d4b5248fc6d5ff1fc927dd3b65. Its review commit is adf1256cff9cde365246a3db2dac90c72fda3b13. The separately frozen primary and signed-sensitivity executions are inherited from 7f4de488134e93b6cfb5dd95a3e6d7e14b45ac2a. This manuscript integration does not rerun those production services or add cost samples. The source freezes, interrupted controller, corrected controller, fitted parameters, raw endpoints, clocks and immutable process checkpoints remain unchanged.

The publication audit reloads each saved fitted critic, reconstructs its proposal, and regenerates each unique whole-domain certificate with the frozen numerical algorithms. Every saved candidate endpoint, covering endpoint, residual band, value band and policy choice is compared with that calculation. Repetition-equivalent inputs are computed once and reused only after their canonical input identities match. This is primitive re-evaluation of the recorded certificate with the same

algorithm, not an independent proof assistant or an independent implementation of the entire economic simulator.

All 132 services are then checked for their exact incumbent transfer, nine common candidates and tenth proposal, complete continuous-action cover, strict minimum-upper selection, feasibility, exact finite-sweep recurrence, declared look order, cumulative sample prefixes and first legal stopping decision. Every referenced file and process-log hash is checked. No later inference or construction is permitted after a target crossing. A finite budget exhaustion remains in the outcome table.

1.2. *Cross-platform moment verification*

Different numerical-library versions can differ in the final rounding of a sample second moment even when their endpoint arrays are byte identical. The audit therefore does not overwrite a scientific interval with a newly rounded interval and call it the same record. Instead, it converts each binary64 endpoint to an exact rational, computes its exact sample mean, second moment and unbiased variance, and checks their containment in the originally stored outward moment account. It reconstructs the rational empirical-Bernstein endpoints from those stored accounts. Thus a valid original interval can be verified without assuming bit identity of a library reduction on another platform.

Validation confidence boxes receive the corresponding exact-mean check, with their stored confidence radius and numerical-bin allowance. All policy-cost endpoints retain signed path gains. Conditional expected-cost safety does not imply that a common-shock pathwise gain is nonnegative; such paths are not clipped away. The declared independent-bin statistical model remains distinct from the reproducibility of a pseudorandom seed.

1.3. *Residual cache and signed tubes*

A range query includes every closed leaf meeting the query box. This is conservative at a boundary but preserves the half-open deployed observation rule. No smoothness is imposed on the actor. The first-stage installed policy is globally zero. Sensitivity is used only under that verified reference identity; a changed incumbent

falls back to the generic cache. The forward zero-policy tube contains the entire future innovation support and the backward tube uses the sparse Jacobian of the original drift. Its first innovation is integrated over the declared bins with exact probabilities. The algorithm does not recursively enumerate all q^T future bins.

Each raw candidate and covering comparison retains its base-cache interval, signed-tube interval and intersection. Empty intersections are errors, not opportunities to choose a midpoint. Table entries for $U - C$ and $C - L$ are maxima of independently nonnegative cellwise quantities. Their separate maxima need not sum to the displayed maximum of $U - L$. Refinement attribution is a comparison of valid enclosures, not a causal decomposition of state discretization, action search, stochastic integration and floating-point error.

1.4. *Exact neural action search*

The new scalar-action solver reads the exact stored dyadic coefficients. It enumerates rational hinge breakpoints, constructs the cubic derivative of each quartic piece, isolates every real stationary root by rational intervals, and compares candidate lattice actions by exact rational objective values. Multiple roots and zero innovation projections have explicit tests. The retained saved-model regressions specify one acquired-cell center in each task before checking its entire finite lattice exhaustively. They create neither a replacement production policy nor a new economic observation. Root-isolation cost and coefficient bit lengths are not replaced by a unit-cost count in a runtime claim.

2. COMPLETE PROSPECTIVE-SERVICE REGISTRY

TABLE I: Every prospective service and timing repetition

Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)
P 2 2	W	49/50	0	1	16384	H	1.149		
P 2 2	W	49/50	1	1	16384	H	1.144		
P 2 2	W	49/50	2	1	16384	H	1.153		
P 2 2	W	9/10	0	2	65536	H	4.664		

	Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)	
1											1
2											2
3		P 2	2	W	9/10	1	2	65536	H	4.617	3
4		P 2	2	W	9/10	2	2	65536	H	4.621	4
5		P 2	2	E	49/50	0	1	16384	H	2.719	5
6		P 2	2	E	49/50	1	1	16384	H	2.714	6
7		P 2	2	E	49/50	2	1	16384	H	2.723	7
8		P 2	2	E	9/10	0	2	65536	H	10.442	8
9		P 2	2	E	9/10	1	2	65536	H	10.313	9
10		P 2	2	E	9/10	2	2	65536	H	10.247	10
11		P 4	4	E	49/50	0	2	16384	H	37.304	11
12		P 4	4	E	49/50	1	2	16384	H	36.875	12
13		P 4	4	E	49/50	2	2	16384	H	37.219	13
14		P 4	4	E	9/10	0	2	16384	E	35.003	14
15		P 4	4	E	9/10	1	2	16384	E	35.275	15
16		P 4	4	E	9/10	2	2	16384	E	35.054	16
17		P 8	6	E	49/50	0	2	65536	E	47.362	17
18		P 8	6	E	49/50	1	2	65536	E	48.146	18
19		P 8	6	E	49/50	2	2	65536	E	47.742	19
20		P 8	6	E	9/10	0	2	4096	E	43.171	20
21		P 8	6	E	9/10	1	2	4096	E	44.311	21
22		P 8	6	E	9/10	2	2	4096	E	43.425	22
23		P 2	2	Q	49/50	0	1	16384	H	0.936	23
24		P 2	2	Q	49/50	1	1	16384	H	0.933	24
25		P 2	2	Q	49/50	2	1	16384	H	0.928	25
26		P 2	2	Q	9/10	0	2	65536	H	3.900	26
27		P 2	2	Q	9/10	1	2	65536	H	3.904	27
28		P 2	2	Q	9/10	2	2	65536	H	3.879	28
29		P 4	4	Q	49/50	0	2	16384	H	10.739	29
30		P 4	4	Q	49/50	1	2	16384	H	10.567	30
31		P 4	4	Q	49/50	2	2	16384	H	10.715	31
32		P 4	4	Q	9/10	0	2	16384	E	9.153	32
		P 4	4	Q	9/10	1	2	16384	E	8.714	
		P 4	4	Q	9/10	2	2	16384	E	8.462	
		P 8	6	Q	49/50	0	2	65536	E	17.204	

	Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)	
1											1
2											2
3		P 8 6		Q 49/50	1	2	65536	E	16.978		3
4		P 8 6		Q 49/50	2	2	65536	E	17.123		4
5		P 8 6		Q 9/10	0	2	4096	E	13.132		5
6		P 8 6		Q 9/10	1	2	4096	E	13.187		6
7		P 8 6		Q 9/10	2	2	4096	E	13.249		7
8		P 2 2		N 49/50	0	1	16384	H	2.372		8
9		P 2 2		N 49/50	1	1	16384	H	2.376		9
10		P 2 2		N 49/50	2	1	16384	H	2.381		10
11		P 2 2		N 9/10	0	2	65536	H	7.159		11
12		P 2 2		N 9/10	1	2	65536	H	7.114		12
13		P 2 2		N 9/10	2	2	65536	H	7.112		13
14		P 4 4		N 49/50	0	2	16384	H	18.806		14
15		P 4 4		N 49/50	1	2	16384	H	18.506		15
16		P 4 4		N 49/50	2	2	16384	H	18.913		16
17		P 4 4		N 9/10	0	2	16384	E	17.163		17
18		P 4 4		N 9/10	1	2	16384	E	17.608		18
19		P 4 4		N 9/10	2	2	16384	E	17.196		19
20		P 8 6		N 49/50	0	2	65536	E	23.934		20
21		P 8 6		N 49/50	1	2	65536	E	24.386		21
22		P 8 6		N 49/50	2	2	65536	E	23.957		22
23		P 8 6		N 9/10	0	2	4096	E	19.988		23
24		P 8 6		N 9/10	1	2	4096	E	20.054		24
25		P 8 6		N 9/10	2	2	4096	E	19.783		25
26		P 2 2		F 49/50	0	1	16384	H	1.202		26
27		P 2 2		F 49/50	1	1	16384	H	1.195		27
28		P 2 2		F 49/50	2	1	16384	H	1.195		28
29		P 2 2		F 9/10	0	2	65536	H	4.761		29
30		P 2 2		F 9/10	1	2	65536	H	4.729		30
31		P 2 2		F 9/10	2	2	65536	H	4.708		31
32		S 2 2		W 49/50	0	1	4096	H	0.727		32
		S 2 2		W 49/50	1	1	4096	H	0.739		
		S 2 2		W 49/50	2	1	4096	H	0.888		
		S 2 2		W 9/10	0	1	16384	H	0.808		

	Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)		
1											1	
2											2	
3		S	2	2	W	9/10	1	1	16384	H	0.781	3
4		S	2	2	W	9/10	2	1	16384	H	1.345	4
5		S	2	2	E	49/50	0	1	4096	H	1.620	5
6		S	2	2	E	49/50	1	1	4096	H	1.865	6
7		S	2	2	E	49/50	2	1	4096	H	1.688	7
8		S	2	2	E	9/10	0	1	16384	H	1.740	8
9		S	2	2	E	9/10	1	1	16384	H	1.678	9
10		S	2	2	E	9/10	2	1	16384	H	1.737	10
11		S	4	4	E	49/50	0	1	4096	H	11.605	11
12		S	4	4	E	49/50	1	1	4096	H	11.700	12
13		S	4	4	E	49/50	2	1	4096	H	11.589	13
14		S	4	4	E	9/10	0	1	4096	H	11.781	14
15		S	4	4	E	9/10	1	1	4096	H	11.515	15
16		S	4	4	E	9/10	2	1	4096	H	11.684	16
17		S	8	6	E	49/50	0	1	4096	H	31.428	17
18		S	8	6	E	49/50	1	1	4096	H	31.032	18
19		S	8	6	E	49/50	2	1	4096	H	31.304	19
20		S	8	6	E	9/10	0	1	16384	H	31.833	20
21		S	8	6	E	9/10	1	1	16384	H	31.886	21
22		S	8	6	E	9/10	2	1	16384	H	35.303	22
23		S	2	2	Q	49/50	0	1	4096	H	0.619	23
24		S	2	2	Q	49/50	1	1	4096	H	0.665	24
25		S	2	2	Q	49/50	2	1	4096	H	0.660	25
26		S	2	2	Q	9/10	0	1	16384	H	0.695	26
27		S	2	2	Q	9/10	1	1	16384	H	0.701	27
28		S	2	2	Q	9/10	2	1	16384	H	0.745	28
29		S	4	4	Q	49/50	0	1	4096	H	5.539	29
30		S	4	4	Q	49/50	1	1	4096	H	5.559	30
31		S	4	4	Q	49/50	2	1	4096	H	5.619	31
32		S	4	4	Q	9/10	0	1	4096	H	5.582	32
		S	4	4	Q	9/10	1	1	4096	H	5.576	
		S	4	4	Q	9/10	2	1	4096	H	5.583	
		S	8	6	Q	49/50	0	1	4096	H	20.344	

Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)
S	8	6	Q	49/50	1	1	4096	H	20.383
S	8	6	Q	49/50	2	1	4096	H	21.123
S	8	6	Q	9/10	0	1	16384	H	21.243
S	8	6	Q	9/10	1	1	16384	H	21.027
S	8	6	Q	9/10	2	1	16384	H	21.882
S	2	2	N	49/50	0	1	4096	H	1.304
S	2	2	N	49/50	1	1	4096	H	1.333
S	2	2	N	49/50	2	1	4096	H	1.292
S	2	2	N	9/10	0	1	16384	H	1.431
S	2	2	N	9/10	1	1	16384	H	1.389
S	2	2	N	9/10	2	1	16384	H	1.409
S	4	4	N	49/50	0	1	4096	H	9.111
S	4	4	N	49/50	1	1	4096	H	9.106
S	4	4	N	49/50	2	1	4096	H	9.140
S	4	4	N	9/10	0	1	4096	H	9.238
S	4	4	N	9/10	1	1	4096	H	9.087
S	4	4	N	9/10	2	1	4096	H	9.073
S	8	6	N	49/50	0	1	4096	H	25.945
S	8	6	N	49/50	1	1	4096	H	26.944
S	8	6	N	49/50	2	1	4096	H	25.865
S	8	6	N	9/10	0	1	16384	H	26.714
S	8	6	N	9/10	1	1	16384	H	30.501
S	8	6	N	9/10	2	1	16384	H	26.771
S	2	2	F	49/50	0	1	4096	H	0.780
S	2	2	F	49/50	1	1	4096	H	0.785
S	2	2	F	49/50	2	1	4096	H	0.747
S	2	2	F	9/10	0	1	16384	H	0.807
S	2	2	F	9/10	1	1	16384	H	0.809
S	2	2	F	9/10	2	1	16384	H	0.853

P: primary cache; S: signed reference extension. H: target attained; E: declared budget exhausted. Repetitions 0–2 have identical training and inference seeds, not independent

training draws. All 132 processes and their immutable cumulative driver checkpoints are included. The full JSON retains every per-stage construction, cache, gate and inference clock.

3. ALL FINAL ECONOMIC INTERVALS

TABLE II: Every final actual-cost, gain and target interval

Cohort	d	T	Method	q	Cost	Gain	Target contrast
P	2	2	W	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	W	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	2	2	E	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	E	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	2	2	Q	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	Q	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	2	2	N	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	N	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	2	2	F	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	F	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	4	4	E	49/50	[1.138, 1.202]	[0.036, 0.102]	[-0.077, -0.011]
P	4	4	E	9/10	[1.138, 1.201]	[0.035, 0.101]	[0.023, 0.088]
P	4	4	Q	49/50	[1.138, 1.202]	[0.036, 0.102]	[-0.077, -0.011]
P	4	4	Q	9/10	[1.138, 1.201]	[0.035, 0.101]	[0.023, 0.088]
P	4	4	N	49/50	[1.138, 1.202]	[0.036, 0.102]	[-0.077, -0.011]
P	4	4	N	9/10	[1.138, 1.201]	[0.035, 0.101]	[0.023, 0.088]
P	8	6	E	49/50	[1.690, 1.711]	[-0.010, 0.010]	[0.024, 0.044]
P	8	6	E	9/10	[1.607, 1.808]	[-0.157, 0.157]	[0.019, 0.323]
P	8	6	Q	49/50	[1.690, 1.711]	[-0.010, 0.010]	[0.024, 0.044]
P	8	6	Q	9/10	[1.607, 1.808]	[-0.157, 0.157]	[0.019, 0.323]
P	8	6	N	49/50	[1.690, 1.711]	[-0.010, 0.010]	[0.024, 0.044]
P	8	6	N	9/10	[1.607, 1.808]	[-0.157, 0.157]	[0.019, 0.323]
S	2	2	W	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.224, -0.048]
S	2	2	W	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]
S	2	2	E	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.225, -0.048]
S	2	2	E	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]

Cohort	d	T	Method	q	Cost	Gain	Target contrast
S	2	2	Q	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.225, -0.048]
S	2	2	Q	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]
S	2	2	N	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.225, -0.048]
S	2	2	N	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]
S	2	2	F	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.225, -0.048]
S	2	2	F	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]
S	4	4	E	49/50	[0.897, 1.066]	[0.119, 0.380]	[-0.354, -0.097]
S	4	4	E	9/10	[0.897, 1.066]	[0.117, 0.378]	[-0.247, -0.003]
S	4	4	Q	49/50	[0.897, 1.066]	[0.119, 0.380]	[-0.354, -0.097]
S	4	4	Q	9/10	[0.897, 1.066]	[0.117, 0.378]	[-0.247, -0.003]
S	4	4	N	49/50	[0.897, 1.066]	[0.119, 0.380]	[-0.354, -0.097]
S	4	4	N	9/10	[0.897, 1.066]	[0.117, 0.378]	[-0.247, -0.003]
S	8	6	E	49/50	[1.342, 1.536]	[0.091, 0.416]	[-0.380, -0.059]
S	8	6	E	9/10	[1.421, 1.479]	[0.214, 0.298]	[-0.125, -0.046]
S	8	6	Q	49/50	[1.342, 1.536]	[0.091, 0.416]	[-0.380, -0.059]
S	8	6	Q	9/10	[1.421, 1.479]	[0.214, 0.298]	[-0.125, -0.046]
S	8	6	N	49/50	[1.342, 1.536]	[0.091, 0.416]	[-0.380, -0.059]
S	8	6	N	9/10	[1.421, 1.479]	[0.214, 0.298]	[-0.125, -0.046]

Intervals are rounded outward and correspond to repetition zero; exact identities across repetitions are separately checked. A negative target-contrast upper endpoint licenses the stated prospective return. Gain intervals use paired path endpoints rather than subtraction of marginal cost intervals. Every intermediate look remains in the service archive.

4. ALL DATEWISE ERROR ACCOUNTS

TABLE III: All datewise directed gaps and enclosure components

Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$
P	2	2	W	49/50	0	0	0	2.195	0.118	2.230

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2		P 2	2	W	49/50	0	1	27	0.607	0.054	0.660	2
3		P 2	2	W	9/10	0	0	0	2.195	0.118	2.230	3
4		P 2	2	W	9/10	0	1	27	0.607	0.054	0.660	4
5		P 2	2	W	9/10	1	0	0	1.481	0.033	1.486	5
6		P 2	2	W	9/10	1	1	93	0.344	0.033	0.377	6
7		P 2	2	E	49/50	0	0	0	2.195	0.118	2.230	7
8		P 2	2	E	49/50	0	1	27	0.607	0.054	0.660	8
9		P 2	2	E	9/10	0	0	0	2.195	0.118	2.230	9
10		P 2	2	E	9/10	0	1	27	0.607	0.054	0.660	10
11		P 2	2	E	9/10	1	0	0	1.481	0.033	1.486	11
12		P 2	2	E	9/10	1	1	93	0.344	0.033	0.377	12
13		P 4	4	E	49/50	0	0	0	4.727	0.062	4.729	13
14		P 4	4	E	49/50	0	1	0	3.705	0.058	3.706	14
15		P 4	4	E	49/50	0	2	0	2.614	0.054	2.615	15
16		P 4	4	E	49/50	0	3	15	0.638	0.054	0.691	16
17		P 4	4	E	49/50	1	0	0	4.530	0.041	4.530	17
18		P 4	4	E	49/50	1	1	0	3.535	0.038	3.536	18
19		P 4	4	E	49/50	1	2	0	2.474	0.035	2.474	19
20		P 4	4	E	49/50	1	3	90	0.636	0.043	0.679	20
21		P 4	4	E	9/10	0	0	0	4.727	0.062	4.729	21
22		P 4	4	E	9/10	0	1	0	3.705	0.058	3.706	22
23		P 4	4	E	9/10	0	2	0	2.614	0.054	2.615	23
24		P 4	4	E	9/10	0	3	15	0.638	0.054	0.691	24
25		P 4	4	E	9/10	1	0	0	4.530	0.041	4.530	25
26		P 4	4	E	9/10	1	1	0	3.535	0.038	3.536	26
27		P 4	4	E	9/10	1	2	0	2.474	0.035	2.474	27
28		P 4	4	E	9/10	1	3	90	0.636	0.043	0.679	28
29		P 8	6	E	49/50	0	0	0	7.047	0.021	7.047	29
30		P 8	6	E	49/50	0	1	0	6.079	0.023	6.079	30
31		P 8	6	E	49/50	0	2	0	5.047	0.024	5.047	31
32		P 8	6	E	49/50	0	3	0	3.946	0.027	3.946	32
		P 8	6	E	49/50	0	4	0	2.771	0.029	2.772	
		P 8	6	E	49/50	0	5	0	0.640	0.074	0.713	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2		P 8 6		E 49/50	1	0		0	6.836	0.058	6.836	2
3		P 8 6		E 49/50	1	1		0	5.901	0.056	5.902	3
4		P 8 6		E 49/50	1	2		0	4.904	0.055	4.904	4
5		P 8 6		E 49/50	1	3		0	3.840	0.053	3.841	5
6		P 8 6		E 49/50	1	4		0	2.705	0.052	2.706	6
7		P 8 6		E 49/50	1	5		0	0.639	0.064	0.702	7
8		P 8 6		E 9/10	0	0		0	7.047	0.021	7.047	8
9		P 8 6		E 9/10	0	1		0	6.079	0.023	6.079	9
10		P 8 6		E 9/10	0	2		0	5.047	0.024	5.047	10
11		P 8 6		E 9/10	0	3		0	3.946	0.027	3.946	11
12		P 8 6		E 9/10	0	4		0	2.771	0.029	2.772	12
13		P 8 6		E 9/10	0	5		0	0.640	0.074	0.713	13
14		P 8 6		E 9/10	1	0		0	6.836	0.058	6.836	14
15		P 8 6		E 9/10	1	1		0	5.901	0.056	5.902	15
16		P 8 6		E 9/10	1	2		0	4.904	0.055	4.904	16
17		P 8 6		E 9/10	1	3		0	3.840	0.053	3.841	17
18		P 8 6		E 9/10	1	4		0	2.705	0.052	2.706	18
19		P 8 6		E 9/10	1	5		0	0.639	0.064	0.702	19
20		P 2 2		Q 49/50	0	0		0	2.195	0.118	2.230	20
21		P 2 2		Q 49/50	0	1		27	0.607	0.054	0.660	21
22		P 2 2		Q 9/10	0	0		0	2.195	0.118	2.230	22
23		P 2 2		Q 9/10	0	1		27	0.607	0.054	0.660	23
24		P 2 2		Q 9/10	1	0		0	1.361	0.033	1.388	24
25		P 2 2		Q 9/10	1	1		93	0.344	0.033	0.377	25
26		P 4 4		Q 49/50	0	0		0	4.727	0.062	4.729	26
27		P 4 4		Q 49/50	0	1		0	3.705	0.058	3.706	27
28		P 4 4		Q 49/50	0	2		0	2.614	0.054	2.615	28
29		P 4 4		Q 49/50	0	3		15	0.638	0.054	0.691	29
30		P 4 4		Q 49/50	1	0		0	4.530	0.041	4.530	30
31		P 4 4		Q 49/50	1	1		0	3.535	0.038	3.536	31
32		P 4 4		Q 49/50	1	2		0	2.474	0.035	2.474	32
		P 4 4		Q 49/50	1	3		90	0.636	0.043	0.679	
		P 4 4		Q 9/10	0	0		0	4.727	0.062	4.729	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2		P 4	4	Q	9/10	0	1	0	3.705	0.058	3.706	2
3		P 4	4	Q	9/10	0	2	0	2.614	0.054	2.615	3
4		P 4	4	Q	9/10	0	3	15	0.638	0.054	0.691	4
5		P 4	4	Q	9/10	1	0	0	4.530	0.041	4.530	5
6		P 4	4	Q	9/10	1	1	0	3.535	0.038	3.536	6
7		P 4	4	Q	9/10	1	2	0	2.474	0.035	2.474	7
8		P 4	4	Q	9/10	1	3	90	0.636	0.043	0.679	8
9		P 8	6	Q	49/50	0	0	0	7.047	0.021	7.047	9
10		P 8	6	Q	49/50	0	1	0	6.079	0.023	6.079	10
11		P 8	6	Q	49/50	0	2	0	5.047	0.024	5.047	11
12		P 8	6	Q	49/50	0	3	0	3.946	0.027	3.946	12
13		P 8	6	Q	49/50	0	4	0	2.771	0.029	2.772	13
14		P 8	6	Q	49/50	0	5	0	0.640	0.074	0.713	14
15		P 8	6	Q	49/50	1	0	0	6.836	0.058	6.836	15
16		P 8	6	Q	49/50	1	1	0	5.901	0.056	5.902	16
17		P 8	6	Q	49/50	1	2	0	4.904	0.055	4.904	17
18		P 8	6	Q	49/50	1	3	0	3.840	0.053	3.841	18
19		P 8	6	Q	49/50	1	4	0	2.705	0.052	2.706	19
20		P 8	6	Q	49/50	1	5	0	0.639	0.064	0.702	20
21		P 8	6	Q	9/10	0	0	0	7.047	0.021	7.047	21
22		P 8	6	Q	9/10	0	1	0	6.079	0.023	6.079	22
23		P 8	6	Q	9/10	0	2	0	5.047	0.024	5.047	23
24		P 8	6	Q	9/10	0	3	0	3.946	0.027	3.946	24
25		P 8	6	Q	9/10	0	4	0	2.771	0.029	2.772	25
26		P 8	6	Q	9/10	0	5	0	0.640	0.074	0.713	26
27		P 8	6	Q	9/10	1	0	0	6.836	0.058	6.836	27
28		P 8	6	Q	9/10	1	1	0	5.901	0.056	5.902	28
29		P 8	6	Q	9/10	1	2	0	4.904	0.055	4.904	29
30		P 8	6	Q	9/10	1	3	0	3.840	0.053	3.841	30
31		P 8	6	Q	9/10	1	4	0	2.705	0.052	2.706	31
32		P 8	6	Q	9/10	1	5	0	0.639	0.064	0.702	32
		P 2	2	N	49/50	0	0	0	2.195	0.118	2.230	
		P 2	2	N	49/50	0	1	27	0.607	0.054	0.660	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2	P	2	2	N	9/10	0	0	0	2.195	0.118	2.230	2
3	P	2	2	N	9/10	0	1	27	0.607	0.054	0.660	3
4	P	2	2	N	9/10	1	0	0	1.481	0.033	1.486	4
5	P	2	2	N	9/10	1	1	93	0.344	0.033	0.377	5
6	P	4	4	N	49/50	0	0	0	4.727	0.062	4.729	6
7	P	4	4	N	49/50	0	1	0	3.705	0.058	3.706	7
8	P	4	4	N	49/50	0	2	0	2.614	0.054	2.615	8
9	P	4	4	N	49/50	0	3	15	0.638	0.054	0.691	9
10	P	4	4	N	49/50	1	0	0	4.530	0.041	4.530	10
11	P	4	4	N	49/50	1	1	0	3.535	0.038	3.536	11
12	P	4	4	N	49/50	1	2	0	2.474	0.035	2.474	12
13	P	4	4	N	49/50	1	3	90	0.636	0.043	0.679	13
14	P	4	4	N	9/10	0	0	0	4.727	0.062	4.729	14
15	P	4	4	N	9/10	0	1	0	3.705	0.058	3.706	15
16	P	4	4	N	9/10	0	2	0	2.614	0.054	2.615	16
17	P	4	4	N	9/10	0	3	15	0.638	0.054	0.691	17
18	P	4	4	N	9/10	1	0	0	4.530	0.041	4.530	18
19	P	4	4	N	9/10	1	1	0	3.535	0.038	3.536	19
20	P	4	4	N	9/10	1	2	0	2.474	0.035	2.474	20
21	P	4	4	N	9/10	1	3	90	0.636	0.043	0.679	21
22	P	8	6	N	49/50	0	0	0	7.047	0.021	7.047	22
23	P	8	6	N	49/50	0	1	0	6.079	0.023	6.079	23
24	P	8	6	N	49/50	0	2	0	5.047	0.024	5.047	24
25	P	8	6	N	49/50	0	3	0	3.946	0.027	3.946	25
26	P	8	6	N	49/50	0	4	0	2.771	0.029	2.772	26
27	P	8	6	N	49/50	0	5	0	0.640	0.074	0.713	27
28	P	8	6	N	49/50	1	0	0	6.836	0.058	6.836	28
29	P	8	6	N	49/50	1	1	0	5.901	0.056	5.902	29
30	P	8	6	N	49/50	1	2	0	4.904	0.055	4.904	30
31	P	8	6	N	49/50	1	3	0	3.840	0.053	3.841	31
32	P	8	6	N	49/50	1	4	0	2.705	0.052	2.706	32
	P	8	6	N	49/50	1	5	0	0.639	0.064	0.702	
	P	8	6	N	9/10	0	0	0	7.047	0.021	7.047	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$		
1												1	
2		P	8	6	N	9/10	0	1	0	6.079	0.023	6.079	2
3		P	8	6	N	9/10	0	2	0	5.047	0.024	5.047	3
4		P	8	6	N	9/10	0	3	0	3.946	0.027	3.946	4
5		P	8	6	N	9/10	0	4	0	2.771	0.029	2.772	5
6		P	8	6	N	9/10	0	5	0	0.640	0.074	0.713	6
7		P	8	6	N	9/10	1	0	0	6.836	0.058	6.836	7
8		P	8	6	N	9/10	1	1	0	5.901	0.056	5.902	8
9		P	8	6	N	9/10	1	2	0	4.904	0.055	4.904	9
10		P	8	6	N	9/10	1	3	0	3.840	0.053	3.841	10
11		P	8	6	N	9/10	1	4	0	2.705	0.052	2.706	11
12		P	8	6	N	9/10	1	5	0	0.639	0.064	0.702	12
13		P	2	2	F	49/50	0	0	0	2.195	0.118	2.230	13
14		P	2	2	F	49/50	0	1	27	0.607	0.054	0.660	14
15		P	2	2	F	9/10	0	0	0	2.195	0.118	2.230	15
16		P	2	2	F	9/10	0	1	27	0.607	0.054	0.660	16
17		P	2	2	F	9/10	1	0	0	1.481	0.033	1.486	17
18		P	2	2	F	9/10	1	1	93	0.344	0.033	0.377	18
19		S	2	2	W	49/50	0	0	30	0.319	0.086	0.405	19
20		S	2	2	W	49/50	0	1	27	0.607	0.054	0.660	20
21		S	2	2	W	9/10	0	0	30	0.319	0.086	0.405	21
22		S	2	2	W	9/10	0	1	27	0.607	0.054	0.660	22
23		S	2	2	E	49/50	0	0	30	0.319	0.086	0.405	23
24		S	2	2	E	49/50	0	1	27	0.607	0.054	0.660	24
25		S	2	2	E	9/10	0	0	30	0.319	0.086	0.405	25
26		S	2	2	E	9/10	0	1	27	0.607	0.054	0.660	26
27		S	4	4	E	49/50	0	0	31	0.503	0.246	0.749	27
28		S	4	4	E	49/50	0	1	31	0.464	0.227	0.691	28
29		S	4	4	E	49/50	0	2	28	0.400	0.197	0.596	29
30		S	4	4	E	49/50	0	3	15	0.638	0.054	0.691	30
31		S	4	4	E	9/10	0	0	31	0.503	0.246	0.749	31
32		S	4	4	E	9/10	0	1	31	0.464	0.227	0.691	32
		S	4	4	E	9/10	0	2	28	0.400	0.197	0.596	
		S	4	4	E	9/10	0	3	15	0.638	0.054	0.691	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2												2
3		S 8 6		E 49/50	0	0		26	0.604	0.418	1.021	3
4		S 8 6		E 49/50	0	1		24	0.588	0.407	0.994	4
5		S 8 6		E 49/50	0	2		24	0.562	0.388	0.950	5
6		S 8 6		E 49/50	0	3		20	0.521	0.358	0.879	6
7		S 8 6		E 49/50	0	4		14	0.453	0.310	0.762	7
8		S 8 6		E 49/50	0	5		0	0.640	0.074	0.713	8
9		S 8 6		E 9/10	0	0		26	0.604	0.418	1.021	9
10		S 8 6		E 9/10	0	1		24	0.588	0.407	0.994	10
11		S 8 6		E 9/10	0	2		24	0.562	0.388	0.950	11
12		S 8 6		E 9/10	0	3		20	0.521	0.358	0.879	12
13		S 8 6		E 9/10	0	4		14	0.453	0.310	0.762	13
14		S 8 6		E 9/10	0	5		0	0.640	0.074	0.713	14
15		S 2 2		Q 49/50	0	0		30	0.319	0.086	0.405	15
16		S 2 2		Q 49/50	0	1		27	0.607	0.054	0.660	16
17		S 2 2		Q 9/10	0	0		30	0.319	0.086	0.405	17
18		S 2 2		Q 9/10	0	1		27	0.607	0.054	0.660	18
19		S 4 4		Q 49/50	0	0		31	0.503	0.246	0.749	19
20		S 4 4		Q 49/50	0	1		31	0.464	0.227	0.691	20
21		S 4 4		Q 49/50	0	2		28	0.400	0.197	0.596	21
22		S 4 4		Q 49/50	0	3		15	0.638	0.054	0.691	22
23		S 4 4		Q 9/10	0	0		31	0.503	0.246	0.749	23
24		S 4 4		Q 9/10	0	1		31	0.464	0.227	0.691	24
25		S 4 4		Q 9/10	0	2		28	0.400	0.197	0.596	25
26		S 4 4		Q 9/10	0	3		15	0.638	0.054	0.691	26
27		S 8 6		Q 49/50	0	0		26	0.604	0.418	1.021	27
28		S 8 6		Q 49/50	0	1		24	0.588	0.407	0.994	28
29		S 8 6		Q 49/50	0	2		24	0.562	0.388	0.950	29
30		S 8 6		Q 49/50	0	3		20	0.521	0.358	0.879	30
31		S 8 6		Q 49/50	0	4		14	0.453	0.310	0.762	31
32		S 8 6		Q 49/50	0	5		0	0.640	0.074	0.713	32
		S 8 6		Q 9/10	0	0		26	0.604	0.418	1.021	
		S 8 6		Q 9/10	0	1		24	0.588	0.407	0.994	
		S 8 6		Q 9/10	0	2		24	0.562	0.388	0.950	

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All stages of repetition zero are retained. A maximum of sums need not equal the sum of maxima. Quantities are upper bounds rounded upward, not estimates of disjoint causal errors. Complete cell arrays contain the exact endpoints used by the gate.

5. READING AND PRESERVATION MAP

The active article is organized around construction and training, acquired-state improvement, prospective stopping and actual economic cost. The preceding main article is retained as an unchanged historical source and a rebuilt development document. The complete development article and complete proof supplement retain the full prior body and add the current theoretical and numerical sections. Historical labels and immutable section-source hashes are checked rather than relying on a statement that no results were removed.

The original resource-allocation, original-owner compiler, observation-price, recursive-preference, diffusion, temporal-self and game results remain available in those companions. The expected-cost sampling and linear action-kernel results in the active article do not claim that a nonlinear recursive utility or game has been experimentally validated by an investment-policy interval. A reading map is a distinction among assumptions and evidence, not abandonment of the original NBO research object.

6. EXECUTED ACTION-SEARCH SERVICES AND COMPLETE NUMERICAL ACCOUNTS

This section records the new R57 execution, separately from the preceding R55 cohorts and their R56 integration. Its scientific protocol, executable source and declared target family were frozen before production. The publication replay does not retrain the critics, choose new seeds, draw new economic observations or replace the original service clocks.

6.1. *Exact search and proposal identity*

For each saved critic, the replay reconstructs the integrated action objective from the exact stored dyadic coefficients. The action-dependent objective drops only a common additive constant. It recomputes rational hinge breakpoints, cubic stationary-root isolation and exact rational comparisons on the action lattice. It

checks the chosen index, objective value, nonterminal menu optimization loss and root/piece counts. Different valid rational isolating brackets are not by themselves different economic actions; the selected lattice index and exact objective are the controlling solver identities.

The ReLU-exact mode keeps every ReLU-menu candidate. The quadratic mode has its own exact fitted-value search; the other modes retain their declared menu construction. Every terminal proposal uses the same analytic terminal objective under the original law. The whole-cell signed reference verifier is independent of the critic, so a shared action receives identical endpoints in the matched search comparison. The replay reconstructs all raw candidate arrays, selected actions, continuous-action lower covers and datewise gaps. Removing the added algebraic action uses the stored pre-addition minimum and policy; no different training run is substituted as the counterfactual menu.

6.2. Every target-service return

TABLE V: Every prospective target-service return

Service	Attempts	Leaves	Paths	Work (s)	Reached
2/2-20-C-1	2	64	131072	3.776	yes
2/2-20-C-2	2	64	131072	3.793	yes
2/2-20-C-3	2	64	131072	3.763	yes
2/2-10-C-1	1	32	16384	1.248	yes
2/2-10-C-2	1	32	16384	1.242	yes
2/2-10-C-3	1	32	16384	1.220	yes
4/4-20-C-1	2	64	131072	12.983	yes
4/4-20-C-2	2	64	131072	12.881	yes
4/4-20-C-3	2	64	131072	12.439	yes
4/4-10-C-1	1	32	4096	3.363	yes
4/4-10-C-2	1	32	4096	3.382	yes
4/4-10-C-3	1	32	4096	3.477	yes
8/6-20-C-1	2	64	81920	35.571	no
8/6-20-C-2	2	64	81920	35.049	no

Every prospective target-service return (continued)

Service	Attempts	Leaves	Paths	Work (s)	Reached
8/6-20-C-3	2	64	81920	35.153	no
8/6-10-C-1	1	32	16384	11.290	yes
8/6-10-C-2	1	32	16384	11.470	yes
8/6-10-C-3	1	32	16384	11.337	yes
2/2-20-E-1	2	64	131072	5.202	yes
2/2-20-E-2	2	64	131072	5.168	yes
2/2-20-E-3	2	64	131072	5.185	yes
2/2-10-E-1	1	32	16384	2.075	yes
2/2-10-E-2	1	32	16384	1.954	yes
2/2-10-E-3	1	32	16384	1.928	yes
4/4-20-E-1	2	64	131072	15.872	yes
4/4-20-E-2	2	64	131072	16.513	yes
4/4-20-E-3	2	64	131072	15.984	yes
4/4-10-E-1	1	32	4096	4.665	yes
4/4-10-E-2	1	32	4096	4.583	yes
4/4-10-E-3	1	32	4096	4.626	yes
8/6-20-E-1	2	64	81920	40.704	no
8/6-20-E-2	2	64	81920	40.766	no
8/6-20-E-3	2	64	81920	41.016	no
8/6-10-E-1	1	32	16384	13.589	yes
8/6-10-E-2	1	32	16384	13.204	yes
8/6-10-E-3	1	32	16384	13.115	yes
2/2-20-Q-1	2	64	131072	4.107	yes
2/2-20-Q-2	2	64	131072	4.043	yes
2/2-20-Q-3	2	64	131072	4.083	yes
2/2-10-Q-1	1	32	16384	1.350	yes
2/2-10-Q-2	1	32	16384	1.340	yes
2/2-10-Q-3	1	32	16384	1.324	yes
4/4-20-Q-1	2	64	131072	13.991	yes
4/4-20-Q-2	2	64	131072	13.475	yes
4/4-20-Q-3	2	64	131072	13.780	yes
4/4-10-Q-1	1	32	4096	3.753	yes

Every prospective target-service return (continued)						
Service	Attempts	Leaves	Paths	Work (s)	Reached	
4/4-10-Q-2	1	32	4096	3.698	yes	
4/4-10-Q-3	1	32	4096	3.713	yes	
8/6-20-Q-1	2	64	81920	38.954	no	
8/6-20-Q-2	2	64	81920	38.257	no	
8/6-20-Q-3	2	64	81920	38.682	no	
8/6-10-Q-1	1	32	16384	12.302	yes	
8/6-10-Q-2	1	32	16384	12.451	yes	
8/6-10-Q-3	1	32	16384	12.564	yes	
2/2-20-X-1	2	64	131072	5.238	yes	
2/2-20-X-2	2	64	131072	5.179	yes	
2/2-20-X-3	2	64	131072	5.216	yes	
2/2-10-X-1	1	32	16384	1.700	yes	
2/2-10-X-2	1	32	16384	1.666	yes	
2/2-10-X-3	1	32	16384	1.697	yes	
4/4-20-X-1	2	64	131072	17.786	yes	
4/4-20-X-2	2	64	131072	17.951	yes	
4/4-20-X-3	2	64	131072	18.369	yes	
4/4-10-X-1	1	32	4096	5.272	yes	
4/4-10-X-2	1	32	4096	4.865	yes	
4/4-10-X-3	1	32	4096	5.122	yes	
8/6-20-X-1	2	64	81920	44.183	no	
8/6-20-X-2	2	64	81920	45.418	no	
8/6-20-X-3	2	64	81920	44.189	no	
8/6-10-X-1	1	32	16384	14.165	yes	
8/6-10-X-2	1	32	16384	13.960	yes	
8/6-10-X-3	1	32	16384	14.161	yes	
2/2-20-N-1	2	64	131072	4.511	yes	
2/2-20-N-2	2	64	131072	4.536	yes	
2/2-20-N-3	2	64	131072	4.497	yes	
2/2-10-N-1	1	32	16384	1.471	yes	
2/2-10-N-2	1	32	16384	1.435	yes	
2/2-10-N-3	1	32	16384	1.458	yes	

Every prospective target-service return (continued)

Service	Attempts	Leaves	Paths	Work (s)	Reached
4/4-20-N-1	2	64	131072	15.004	yes
4/4-20-N-2	2	64	131072	15.240	yes
4/4-20-N-3	2	64	131072	15.077	yes
4/4-10-N-1	1	32	4096	3.929	yes
4/4-10-N-2	1	32	4096	3.963	yes
4/4-10-N-3	1	32	4096	3.949	yes
8/6-20-N-1	2	64	81920	39.190	no
8/6-20-N-2	2	64	81920	39.458	no
8/6-20-N-3	2	64	81920	39.239	no
8/6-10-N-1	1	32	16384	12.484	yes
8/6-10-N-2	1	32	16384	12.188	yes
8/6-10-N-3	1	32	16384	12.581	yes

Service identifiers encode dimension/horizon, percent-reduction target, mode, and seed index 1–3.
 Paths sum the final cumulative look in each attempted construction; nested looks are not double-counted.
 All 90 services are retained, including every budget exhaustion.

TABLE VI: Every returned policy and its original Bellman account

Service	Expected cost	Gain from installed	All-state bound
2/2-20-C-1	[0.59687, 0.61854]	[0.15900, 0.17347]	3.85741
2/2-20-C-2	[0.59646, 0.61815]	[0.15944, 0.17391]	3.85741
2/2-20-C-3	[0.59766, 0.61949]	[0.15938, 0.17388]	3.85741
2/2-10-C-1	[0.59456, 0.64992]	[0.12935, 0.17762]	3.94572
2/2-10-C-2	[0.59742, 0.65311]	[0.12966, 0.17792]	3.94572
2/2-10-C-3	[0.59889, 0.65459]	[0.13026, 0.17850]	3.94572
4/4-20-C-1	[0.95342, 0.97249]	[0.26227, 0.28146]	6.72215
4/4-20-C-2	[0.95352, 0.97261]	[0.26187, 0.28108]	6.72215
4/4-20-C-3	[0.95430, 0.97342]	[0.26221, 0.28144]	6.72215
4/4-10-C-1	[0.90014, 1.06847]	[0.11978, 0.37950]	6.78416
4/4-10-C-2	[0.89574, 1.06400]	[0.11750, 0.37711]	6.78416

1	Every returned policy and its original Bellman account (continued)				1
2	Service	Expected cost	Gain from installed	All-state bound	2
3	4/4-10-C-3	[0.90078, 1.06845]	[0.12085, 0.38038]	6.78416	3
4	8/6-20-C-1	[1.34699, 1.36544]	[0.33397, 0.35624]	9.18783	4
5	8/6-20-C-2	[1.34634, 1.36475]	[0.33372, 0.35598]	9.18783	5
6	8/6-20-C-3	[1.34640, 1.36482]	[0.33384, 0.35609]	9.18783	6
7	8/6-10-C-1	[1.41623, 1.47339]	[0.21298, 0.29622]	9.24880	7
8	8/6-10-C-2	[1.42016, 1.47724]	[0.21348, 0.29669]	9.24880	8
9	8/6-10-C-3	[1.41810, 1.47521]	[0.21329, 0.29649]	9.24880	9
10	2/2-20-E-1	[0.59681, 0.61848]	[0.15905, 0.17353]	3.85741	10
11	2/2-20-E-2	[0.59636, 0.61805]	[0.15954, 0.17401]	3.85741	11
12	2/2-20-E-3	[0.59780, 0.61962]	[0.15924, 0.17374]	3.85741	12
13	2/2-10-E-1	[0.59456, 0.64992]	[0.12935, 0.17762]	3.94572	13
14	2/2-10-E-2	[0.59742, 0.65311]	[0.12966, 0.17792]	3.94572	14
15	2/2-10-E-3	[0.59889, 0.65459]	[0.13026, 0.17850]	3.94572	15
16	4/4-20-E-1	[0.95347, 0.97254]	[0.26221, 0.28141]	6.72215	16
17	4/4-20-E-2	[0.95352, 0.97261]	[0.26187, 0.28108]	6.72215	17
18	4/4-20-E-3	[0.95430, 0.97342]	[0.26221, 0.28144]	6.72215	18
19	4/4-10-E-1	[0.90014, 1.06847]	[0.11978, 0.37950]	6.78416	19
20	4/4-10-E-2	[0.89574, 1.06400]	[0.11750, 0.37711]	6.78416	20
21	4/4-10-E-3	[0.90062, 1.06826]	[0.12102, 0.38056]	6.78416	21
22	8/6-20-E-1	[1.34699, 1.36544]	[0.33397, 0.35624]	9.18783	22
23	8/6-20-E-2	[1.34626, 1.36467]	[0.33380, 0.35606]	9.18783	23
24	8/6-20-E-3	[1.34627, 1.36469]	[0.33397, 0.35621]	9.18783	24
25	8/6-10-E-1	[1.41623, 1.47339]	[0.21298, 0.29622]	9.24880	25
26	8/6-10-E-2	[1.42016, 1.47724]	[0.21348, 0.29669]	9.24880	26
27	8/6-10-E-3	[1.41810, 1.47521]	[0.21329, 0.29649]	9.24880	27
28	2/2-20-Q-1	[0.59687, 0.61854]	[0.15900, 0.17347]	3.85741	28
29	2/2-20-Q-2	[0.59646, 0.61815]	[0.15944, 0.17391]	3.85741	29
30	2/2-20-Q-3	[0.59766, 0.61949]	[0.15938, 0.17388]	3.85741	30
31	2/2-10-Q-1	[0.59456, 0.64992]	[0.12935, 0.17762]	3.94572	31
32	2/2-10-Q-2	[0.59742, 0.65311]	[0.12966, 0.17792]	3.94572	32
33	2/2-10-Q-3	[0.59889, 0.65459]	[0.13026, 0.17850]	3.94572	33
34	4/4-20-Q-1	[0.95342, 0.97249]	[0.26227, 0.28146]	6.72215	34

Every returned policy and its original Bellman account (continued)			
Service	Expected cost	Gain from installed	All-state bound
4/4-20-Q-2	[0.95352, 0.97261]	[0.26187, 0.28108]	6.72215
4/4-20-Q-3	[0.95430, 0.97342]	[0.26221, 0.28144]	6.72215
4/4-10-Q-1	[0.90014, 1.06847]	[0.11978, 0.37950]	6.78416
4/4-10-Q-2	[0.89574, 1.06400]	[0.11750, 0.37711]	6.78416
4/4-10-Q-3	[0.90078, 1.06845]	[0.12085, 0.38038]	6.78416
8/6-20-Q-1	[1.34699, 1.36544]	[0.33397, 0.35624]	9.18783
8/6-20-Q-2	[1.34634, 1.36475]	[0.33372, 0.35598]	9.18783
8/6-20-Q-3	[1.34640, 1.36482]	[0.33384, 0.35609]	9.18783
8/6-10-Q-1	[1.41623, 1.47339]	[0.21298, 0.29622]	9.24880
8/6-10-Q-2	[1.42016, 1.47724]	[0.21348, 0.29669]	9.24880
8/6-10-Q-3	[1.41810, 1.47521]	[0.21329, 0.29649]	9.24880
2/2-20-X-1	[0.59688, 0.61856]	[0.15898, 0.17346]	3.85741
2/2-20-X-2	[0.59648, 0.61817]	[0.15942, 0.17389]	3.85741
2/2-20-X-3	[0.59765, 0.61948]	[0.15939, 0.17389]	3.85741
2/2-10-X-1	[0.59456, 0.64992]	[0.12935, 0.17762]	3.94572
2/2-10-X-2	[0.59742, 0.65311]	[0.12966, 0.17792]	3.94572
2/2-10-X-3	[0.59892, 0.65461]	[0.13023, 0.17848]	3.94572
4/4-20-X-1	[0.95335, 0.97242]	[0.26234, 0.28152]	6.72215
4/4-20-X-2	[0.95346, 0.97255]	[0.26193, 0.28114]	6.72215
4/4-20-X-3	[0.95430, 0.97342]	[0.26221, 0.28144]	6.72215
4/4-10-X-1	[0.90014, 1.06847]	[0.11978, 0.37950]	6.78416
4/4-10-X-2	[0.89574, 1.06400]	[0.11750, 0.37711]	6.78416
4/4-10-X-3	[0.90078, 1.06845]	[0.12085, 0.38038]	6.78416
8/6-20-X-1	[1.34699, 1.36544]	[0.33397, 0.35624]	9.18783
8/6-20-X-2	[1.34634, 1.36475]	[0.33372, 0.35598]	9.18783
8/6-20-X-3	[1.34640, 1.36482]	[0.33384, 0.35609]	9.18783
8/6-10-X-1	[1.41623, 1.47339]	[0.21298, 0.29622]	9.24880
8/6-10-X-2	[1.42016, 1.47724]	[0.21348, 0.29669]	9.24880
8/6-10-X-3	[1.41810, 1.47521]	[0.21329, 0.29649]	9.24880
2/2-20-N-1	[0.59687, 0.61854]	[0.15900, 0.17347]	3.85741
2/2-20-N-2	[0.59646, 0.61815]	[0.15944, 0.17391]	3.85741
2/2-20-N-3	[0.59766, 0.61949]	[0.15938, 0.17388]	3.85741

Every returned policy and its original Bellman account (continued)				
Service	Expected cost	Gain from installed	All-state bound	
2/2-10-N-1	[0.59456, 0.64992]	[0.12935, 0.17762]	3.94572	
2/2-10-N-2	[0.59742, 0.65311]	[0.12966, 0.17792]	3.94572	
2/2-10-N-3	[0.59889, 0.65459]	[0.13026, 0.17850]	3.94572	
4/4-20-N-1	[0.95335, 0.97242]	[0.26234, 0.28152]	6.72215	
4/4-20-N-2	[0.95345, 0.97254]	[0.26194, 0.28114]	6.72215	
4/4-20-N-3	[0.95430, 0.97342]	[0.26221, 0.28144]	6.72215	
4/4-10-N-1	[0.90014, 1.06847]	[0.11978, 0.37950]	6.78416	
4/4-10-N-2	[0.89574, 1.06400]	[0.11750, 0.37711]	6.78416	
4/4-10-N-3	[0.90078, 1.06845]	[0.12085, 0.38038]	6.78416	
8/6-20-N-1	[1.34699, 1.36544]	[0.33397, 0.35624]	9.18783	
8/6-20-N-2	[1.34634, 1.36475]	[0.33372, 0.35598]	9.18783	
8/6-20-N-3	[1.34640, 1.36482]	[0.33384, 0.35609]	9.18783	
8/6-10-N-1	[1.41623, 1.47339]	[0.21298, 0.29622]	9.24880	
8/6-10-N-2	[1.42016, 1.47724]	[0.21348, 0.29669]	9.24880	
8/6-10-N-3	[1.41810, 1.47521]	[0.21329, 0.29649]	9.24880	

Intervals are outward rounded. The last column is the one-sweep policy-loss bound against the original Bellman optimum, including the old-continuation term, not merely the local greedy gap. The expected-cost target and this conservative uniform bound are different quantities. Each service is a separate from-primitives process with its own inference and durable output. Within a service, resource refinement restarts from the same installed reference, and no intermediate candidate is deployed. A target-attained return requires a nonpositive confidence upper bound on the signed target estimand. A budget-exhausted return carries a statewise-safe candidate but no attained-target claim. The registry preserves the complete declared set rather than selecting successful seeds.

The all-state column uses the support-capped one-sweep recursion against the original Bellman optimum. It is not the directed greedy gap alone. The old continuation error remains part of the account even when every selected action is verified nonworsening. The exact finite-sweep theorem retains its own premises; two re-

TABLE IV
PROSPECTIVE TEN-PERCENT TARGET SERVICES

d/T	Mode	Reached	Cost upper	Median (s)	Range (s)
2/2	C	3/3	0.65459	1.242	1.220–1.248
2/2	N	3/3	0.65459	1.458	1.435–1.471
2/2	X	3/3	0.65461	1.697	1.666–1.700
2/2	Q	3/3	0.65459	1.340	1.324–1.350
2/2	E	3/3	0.65459	1.954	1.928–2.075
4/4	C	3/3	1.06847	3.382	3.363–3.477
4/4	N	3/3	1.06847	3.949	3.929–3.963
4/4	X	3/3	1.06847	5.122	4.865–5.272
4/4	Q	3/3	1.06847	3.713	3.698–3.753
4/4	E	3/3	1.06847	4.626	4.583–4.665
8/6	C	3/3	1.47724	11.337	11.290–11.470
8/6	N	3/3	1.47724	12.484	12.188–12.581
8/6	X	3/3	1.47724	14.161	13.960–14.165
8/6	Q	3/3	1.47724	12.451	12.302–12.564
8/6	E	3/3	1.47724	13.204	13.115–13.589

C uses common candidates without fitting. N uses the trained ReLU menu; X adds its exact algebraic witness; Q uses quadratic exact search; E uses ExtraTrees menu proposals. All modes receive the same exact terminal action. Three different training seeds are reconstructed separately. Cost upper is the largest returned-policy upper endpoint across those seeds, not a confidence bound for a population mean. Complete process times include every own failed attempt and inference look.

source refinements against a fixed installed reference are not two simultaneous policy-iteration passes against successive incumbents.

6.3. Independent returned-policy comparisons

TABLE VII: Every independent validation contrast

d/T	Target	Seed	X minus	Cost difference	Conclusion
2/2	20	1 C		[-0.00513, 0.00516]	unresolved

	Every independent validation contrast (continued)					
	d/T	Target	Seed	X minus	Cost difference	Conclusion
1	2/2	20	1	E	[-0.00508, 0.00522]	unresolved
2	2/2	20	1	Q	[-0.00513, 0.00516]	unresolved
3	2/2	20	1	N	[-0.00513, 0.00516]	unresolved
4	2/2	20	2	C	[-0.00513, 0.00517]	unresolved
5	2/2	20	2	E	[-0.00504, 0.00528]	unresolved
6	2/2	20	2	Q	[-0.00513, 0.00517]	unresolved
7	2/2	20	2	N	[-0.00513, 0.00516]	unresolved
8	2/2	20	3	C	[-0.00515, 0.00513]	unresolved
9	2/2	20	3	E	[-0.00529, 0.00501]	unresolved
10	2/2	20	3	Q	[-0.00515, 0.00513]	unresolved
11	2/2	20	3	N	[-0.00515, 0.00513]	unresolved
12	2/2	10	1	C	[0.00000, 0.00000]	identity
13	2/2	10	1	E	[0.00000, 0.00000]	identity
14	2/2	10	1	Q	[0.00000, 0.00000]	identity
15	2/2	10	1	N	[0.00000, 0.00000]	identity
16	2/2	10	2	C	[0.00000, 0.00000]	identity
17	2/2	10	2	E	[0.00000, 0.00000]	identity
18	2/2	10	2	Q	[0.00000, 0.00000]	identity
19	2/2	10	2	N	[0.00000, 0.00000]	identity
20	2/2	10	3	C	[-0.00512, 0.00517]	unresolved
21	2/2	10	3	E	[-0.00512, 0.00517]	unresolved
22	2/2	10	3	Q	[-0.00512, 0.00517]	unresolved
23	2/2	10	3	N	[-0.00512, 0.00517]	unresolved
24	4/4	20	1	C	[-0.00772, 0.00758]	unresolved
25	4/4	20	1	E	[-0.00779, 0.00753]	unresolved
26	4/4	20	1	Q	[-0.00772, 0.00758]	unresolved
27	4/4	20	1	N	[0.00000, 0.00000]	identity
28	4/4	20	2	C	[-0.00771, 0.00758]	unresolved
29	4/4	20	2	E	[-0.00771, 0.00758]	unresolved
30	4/4	20	2	Q	[-0.00771, 0.00758]	unresolved
31	4/4	20	2	N	[-0.00763, 0.00764]	unresolved
32	4/4	20	3	C	[-0.00763, 0.00763]	unresolved

Every independent validation contrast (continued)

d/T	Target	Seed	X minus	Cost difference	Conclusion
4/4	20	3 E		[-0.00763, 0.00763]	unresolved
4/4	20	3 Q		[-0.00763, 0.00763]	unresolved
4/4	20	3 N		[-0.00763, 0.00763]	unresolved
4/4	10	1 C		[0.00000, 0.00000]	identity
4/4	10	1 E		[0.00000, 0.00000]	identity
4/4	10	1 Q		[0.00000, 0.00000]	identity
4/4	10	1 N		[0.00000, 0.00000]	identity
4/4	10	2 C		[0.00000, 0.00000]	identity
4/4	10	2 E		[0.00000, 0.00000]	identity
4/4	10	2 Q		[0.00000, 0.00000]	identity
4/4	10	2 N		[0.00000, 0.00000]	identity
4/4	10	3 C		[0.00000, 0.00000]	identity
4/4	10	3 E		[-0.00748, 0.00782]	unresolved
4/4	10	3 Q		[0.00000, 0.00000]	identity
4/4	10	3 N		[0.00000, 0.00000]	identity
8/6	20	1 C		[0.00000, 0.00000]	identity
8/6	20	1 E		[0.00000, 0.00000]	identity
8/6	20	1 Q		[0.00000, 0.00000]	identity
8/6	20	1 N		[0.00000, 0.00000]	identity
8/6	20	2 C		[0.00000, 0.00000]	identity
8/6	20	2 E		[-0.00975, 0.00990]	unresolved
8/6	20	2 Q		[0.00000, 0.00000]	identity
8/6	20	2 N		[0.00000, 0.00000]	identity
8/6	20	3 C		[0.00000, 0.00000]	identity
8/6	20	3 E		[-0.00973, 0.00996]	unresolved
8/6	20	3 Q		[0.00000, 0.00000]	identity
8/6	20	3 N		[0.00000, 0.00000]	identity
8/6	10	1 C		[0.00000, 0.00000]	identity
8/6	10	1 E		[0.00000, 0.00000]	identity
8/6	10	1 Q		[0.00000, 0.00000]	identity
8/6	10	1 N		[0.00000, 0.00000]	identity
8/6	10	2 C		[0.00000, 0.00000]	identity

Every independent validation contrast (continued)

d/T	Target	Seed	X minus	Cost difference	Conclusion
8/6	10	2	E	[0.00000, 0.00000]	identity
8/6	10	2	Q	[0.00000, 0.00000]	identity
8/6	10	2	N	[0.00000, 0.00000]	identity
8/6	10	3	C	[0.00000, 0.00000]	identity
8/6	10	3	E	[0.00000, 0.00000]	identity
8/6	10	3	Q	[0.00000, 0.00000]	identity
8/6	10	3	N	[0.00000, 0.00000]	identity

Each nonidentity interval comes from 65,536 fresh continuous-bin paths for the five returned policies. Negative values favor X. Full raw endpoint arrays, stream hashes, fitted models and prospective stopping records are deposited. The validation stream is generated only after the corresponding target services have returned. The replay reconstructs its original forty-bit initial-state and innovation bins and recomputes every path-cost enclosure. It likewise reconstructs every original stopping stream and every declared look, including unsuccessful early looks. Replaying the same frozen stream is verification, not an increase in sample size.

Original confidence records are verified using exact rational first and second sample moments and the inherited rounded-moment bounds. This separates an arithmetic guarantee from platform-dependent floating-point reduction order. The support for a signed difference includes negative values. Statewise expected safety relative to the installed policy is not used to clip common-shock path gains. Exact policy identities and unresolved statistical differences are retained as different cases.

The new confidence family bounds all declared stopping and paired intervals simultaneously. It is not a claim of independence across methods, targets or reused policy arrays. The three different training seeds generate different fitted objects; they do not establish an optimizer-success distribution or generalization across an unobserved population of economic environments.

6.4. Work, storage and the recorded timing boundary

TABLE VIII: Recorded work components at the twenty-percent target

d/T	Mode	Fit (s)	Search (s)	Verify (s)	Infer (s)	Bytes	RSS (KiB)
2/2	C	0.025	0.124	1.522	1.644	5389008	113044
2/2	N	0.629	0.233	1.525	1.652	5685048	131260
2/2	X	0.626	0.828	1.624	1.659	5729943	130796
2/2	Q	0.095	0.239	1.622	1.660	5706573	113044
2/2	E	1.057	0.325	1.554	1.678	6768970	161412
4/4	C	0.020	0.112	8.649	3.632	5423386	116776
4/4	N	2.417	0.391	8.191	3.584	6345224	133160
4/4	X	2.446	2.705	8.715	3.590	6494162	134948
4/4	Q	0.281	0.471	8.878	3.617	6405642	116776
4/4	E	2.791	0.674	8.257	3.593	8949247	164792
8/6	C	0.018	0.101	30.398	4.264	3531723	117020
8/6	N	3.214	0.530	30.653	4.313	5899194	133432
8/6	X	3.191	4.336	31.978	4.338	6148289	136324
8/6	Q	0.828	0.928	32.146	4.306	5967697	117020
8/6	E	4.192	0.947	30.614	4.281	10024709	168828

Components sum within a run, but their separate medians need not sum to the median complete process time. They exclude some serialization and process overhead that is included in the complete clock. Bytes and RSS are maxima across the three runs. Cross-task runners are not treated as a controlled hardware-scaling regression. The complete own-service clock surrounds the subprocess through termination and durable service records. Internal components separately measure fitting, algebraic search, safety verification and inference through endpoint serialization. Exact rational search pays for its coefficient arithmetic, stationary-root isolation and candidate comparisons. Logical operation counts do not replace the measured process cost, and process seconds are not claimed to be machine independent.

The supplementary paired-comparison timer is narrower: it ends after endpoint computation and serialization, before the final comparison JSON is committed. The driver executes this comparison in its parent process. Thus the intended sep-

arate complete comparison-process timer in the protocol was not obtained. The recorded value is a partial, additional study-overhead measurement, not a complete deployment clock. All own-target-service clocks remain complete. The publication preserves and labels the narrower timing rather than inventing an unobserved overhead or silently amending the raw record.

6.5. Datewise decomposition and preservation

TABLE IX: Datewise verification for the first declared seed

Service	Attempt	Date	Changed	Extra	$U - C$	$C - L$	$U - L$
2/2-20-C-1	1	0	30	0	0.31866	0.08581	0.40446
2/2-20-C-1	1	1	27	0	0.60626	0.05340	0.65965
2/2-20-C-1	2	0	62	0	0.26553	0.05781	0.31615
2/2-20-C-1	2	1	58	0	0.45576	0.04002	0.49577
4/4-20-C-1	1	0	31	0	0.50284	0.24561	0.74845
4/4-20-C-1	1	1	31	0	0.46336	0.22679	0.69015
4/4-20-C-1	1	2	28	0	0.39956	0.19612	0.59568
4/4-20-C-1	1	3	15	0	0.63704	0.05329	0.69033
4/4-20-C-1	2	0	63	0	0.47865	0.20779	0.68644
4/4-20-C-1	2	1	63	0	0.43964	0.19181	0.63144
4/4-20-C-1	2	2	59	0	0.37687	0.16586	0.54273
4/4-20-C-1	2	3	42	0	0.63602	0.04785	0.68386
8/6-20-C-1	1	0	26	0	0.60302	0.41769	1.02070
8/6-20-C-1	1	1	24	0	0.58712	0.40623	0.99334
8/6-20-C-1	1	2	24	0	0.56153	0.38781	0.94934
8/6-20-C-1	1	3	20	0	0.52013	0.35796	0.87809
8/6-20-C-1	1	4	14	0	0.45268	0.30907	0.76175
8/6-20-C-1	1	5	0	0	0.63956	0.07309	0.71264
8/6-20-C-1	2	0	60	0	0.58520	0.37453	0.95973
8/6-20-C-1	2	1	60	0	0.56954	0.36417	0.93371
8/6-20-C-1	2	2	59	0	0.54452	0.34754	0.89205
8/6-20-C-1	2	3	54	0	0.50409	0.32066	0.82475
8/6-20-C-1	2	4	46	0	0.43849	0.27676	0.71525

Datewise verification for the first declared seed (continued)								
Service	Attempt	Date	Changed	Extra	$U - C$	$C - L$	$U - L$	
8/6-20-C-1	2	5	0	0	0.63853	0.06811	0.70663	
2/2-20-E-1	1	0	30	0	0.31866	0.08581	0.40446	
2/2-20-E-1	1	1	27	0	0.60626	0.05340	0.65965	
2/2-20-E-1	2	0	62	0	0.26553	0.05781	0.31615	
2/2-20-E-1	2	1	58	0	0.45576	0.04002	0.49577	
4/4-20-E-1	1	0	31	0	0.50284	0.24561	0.74845	
4/4-20-E-1	1	1	31	0	0.46336	0.22679	0.69015	
4/4-20-E-1	1	2	28	0	0.39956	0.19612	0.59568	
4/4-20-E-1	1	3	15	0	0.63704	0.05329	0.69033	
4/4-20-E-1	2	0	63	0	0.47865	0.20779	0.68644	
4/4-20-E-1	2	1	63	0	0.43964	0.19181	0.63144	
4/4-20-E-1	2	2	59	0	0.37687	0.16586	0.54273	
4/4-20-E-1	2	3	42	0	0.63602	0.04785	0.68386	
8/6-20-E-1	1	0	26	0	0.60302	0.41769	1.02070	
8/6-20-E-1	1	1	24	0	0.58712	0.40623	0.99334	
8/6-20-E-1	1	2	24	0	0.56153	0.38781	0.94934	
8/6-20-E-1	1	3	20	0	0.52013	0.35796	0.87809	
8/6-20-E-1	1	4	14	0	0.45268	0.30907	0.76175	
8/6-20-E-1	1	5	0	0	0.63956	0.07309	0.71264	
8/6-20-E-1	2	0	60	0	0.58520	0.37453	0.95973	
8/6-20-E-1	2	1	60	0	0.56954	0.36417	0.93371	
8/6-20-E-1	2	2	59	0	0.54452	0.34754	0.89205	
8/6-20-E-1	2	3	54	0	0.50409	0.32066	0.82475	
8/6-20-E-1	2	4	46	0	0.43849	0.27676	0.71525	
8/6-20-E-1	2	5	0	0	0.63853	0.06811	0.70663	
2/2-20-Q-1	1	0	30	0	0.31866	0.08581	0.40446	
2/2-20-Q-1	1	1	27	0	0.60626	0.05340	0.65965	
2/2-20-Q-1	2	0	62	0	0.26553	0.05781	0.31615	
2/2-20-Q-1	2	1	58	0	0.45576	0.04002	0.49577	
4/4-20-Q-1	1	0	31	0	0.50284	0.24561	0.74845	
4/4-20-Q-1	1	1	31	0	0.46336	0.22679	0.69015	
4/4-20-Q-1	1	2	28	0	0.39956	0.19612	0.59568	

Datewise verification for the first declared seed (continued)								
Service	Attempt	Date	Changed	Extra	$U - C$	$C - L$	$U - L$	
4/4-20-Q-1	1	3	15	0	0.63704	0.05329	0.69033	
4/4-20-Q-1	2	0	63	0	0.47865	0.20779	0.68644	
4/4-20-Q-1	2	1	63	0	0.43964	0.19181	0.63144	
4/4-20-Q-1	2	2	59	0	0.37687	0.16586	0.54273	
4/4-20-Q-1	2	3	42	0	0.63602	0.04785	0.68386	
8/6-20-Q-1	1	0	26	0	0.60302	0.41769	1.02070	
8/6-20-Q-1	1	1	24	0	0.58712	0.40623	0.99334	
8/6-20-Q-1	1	2	24	0	0.56153	0.38781	0.94934	
8/6-20-Q-1	1	3	20	0	0.52013	0.35796	0.87809	
8/6-20-Q-1	1	4	14	0	0.45268	0.30907	0.76175	
8/6-20-Q-1	1	5	0	0	0.63956	0.07309	0.71264	
8/6-20-Q-1	2	0	60	0	0.58520	0.37453	0.95973	
8/6-20-Q-1	2	1	60	0	0.56954	0.36417	0.93371	
8/6-20-Q-1	2	2	59	0	0.54452	0.34754	0.89205	
8/6-20-Q-1	2	3	54	0	0.50409	0.32066	0.82475	
8/6-20-Q-1	2	4	46	0	0.43849	0.27676	0.71525	
8/6-20-Q-1	2	5	0	0	0.63853	0.06811	0.70663	
2/2-20-X-1	1	0	30	0	0.31866	0.08581	0.40446	
2/2-20-X-1	1	1	27	0	0.60626	0.05340	0.65965	
2/2-20-X-1	2	0	62	1	0.26553	0.05781	0.31615	
2/2-20-X-1	2	1	58	0	0.45576	0.04002	0.49577	
4/4-20-X-1	1	0	31	0	0.50284	0.24561	0.74845	
4/4-20-X-1	1	1	31	0	0.46336	0.22679	0.69015	
4/4-20-X-1	1	2	28	0	0.39956	0.19612	0.59568	
4/4-20-X-1	1	3	15	0	0.63704	0.05329	0.69033	
4/4-20-X-1	2	0	63	0	0.47865	0.20779	0.68644	
4/4-20-X-1	2	1	63	0	0.43964	0.19181	0.63144	
4/4-20-X-1	2	2	59	0	0.37687	0.16586	0.54273	
4/4-20-X-1	2	3	42	0	0.63602	0.04785	0.68386	
8/6-20-X-1	1	0	26	0	0.60302	0.41769	1.02070	
8/6-20-X-1	1	1	24	0	0.58712	0.40623	0.99334	
8/6-20-X-1	1	2	24	0	0.56153	0.38781	0.94934	

Datewise verification for the first declared seed (continued)								
Service	Attempt	Date	Changed	Extra	$U - C$	$C - L$	$U - L$	
8/6-20-X-1	1	3	20	0	0.52013	0.35796	0.87809	
8/6-20-X-1	1	4	14	0	0.45268	0.30907	0.76175	
8/6-20-X-1	1	5	0	0	0.63956	0.07309	0.71264	
8/6-20-X-1	2	0	60	0	0.58520	0.37453	0.95973	
8/6-20-X-1	2	1	60	0	0.56954	0.36417	0.93371	
8/6-20-X-1	2	2	59	0	0.54452	0.34754	0.89205	
8/6-20-X-1	2	3	54	0	0.50409	0.32066	0.82475	
8/6-20-X-1	2	4	46	0	0.43849	0.27676	0.71525	
8/6-20-X-1	2	5	0	0	0.63853	0.06811	0.70663	
2/2-20-N-1	1	0	30	0	0.31866	0.08581	0.40446	
2/2-20-N-1	1	1	27	0	0.60626	0.05340	0.65965	
2/2-20-N-1	2	0	62	0	0.26553	0.05781	0.31615	
2/2-20-N-1	2	1	58	0	0.45576	0.04002	0.49577	
4/4-20-N-1	1	0	31	0	0.50284	0.24561	0.74845	
4/4-20-N-1	1	1	31	0	0.46336	0.22679	0.69015	
4/4-20-N-1	1	2	28	0	0.39956	0.19612	0.59568	
4/4-20-N-1	1	3	15	0	0.63704	0.05329	0.69033	
4/4-20-N-1	2	0	63	0	0.47865	0.20779	0.68644	
4/4-20-N-1	2	1	63	0	0.43964	0.19181	0.63144	
4/4-20-N-1	2	2	59	0	0.37687	0.16586	0.54273	
4/4-20-N-1	2	3	42	0	0.63602	0.04785	0.68386	
8/6-20-N-1	1	0	26	0	0.60302	0.41769	1.02070	
8/6-20-N-1	1	1	24	0	0.58712	0.40623	0.99334	
8/6-20-N-1	1	2	24	0	0.56153	0.38781	0.94934	
8/6-20-N-1	1	3	20	0	0.52013	0.35796	0.87809	
8/6-20-N-1	1	4	14	0	0.45268	0.30907	0.76175	
8/6-20-N-1	1	5	0	0	0.63956	0.07309	0.71264	
8/6-20-N-1	2	0	60	0	0.58520	0.37453	0.95973	
8/6-20-N-1	2	1	60	0	0.56954	0.36417	0.93371	
8/6-20-N-1	2	2	59	0	0.54452	0.34754	0.89205	
8/6-20-N-1	2	3	54	0	0.50409	0.32066	0.82475	
8/6-20-N-1	2	4	46	0	0.43849	0.27676	0.71525	

Datewise verification for the first declared seed (continued)							
Service	Attempt	Date	Changed	Extra	$U - C$	$C - L$	$U - L$
8/6-20-N-1	2	5	0	0	0.63853	0.06811	0.70663

This printed subset uses the first prespecified seed and the twenty-percent target for all modes. Every seed and target is in the machine-readable audit. Each numerical entry is a maximum over leaves; maxima of the first two components need not add to the maximum total gap. The displayed first-seed subset is specified mechanically, not chosen for favorable outcomes. Full datewise arrays for every seed, target and attempted policy remain in the machine-readable replay and raw records. For each leaf the identity $(U - L) = (U - C) + (C - L)$ is exact over the recorded endpoints; printed entries are separate outward-rounded maxima. Acquisition, innovation and arithmetic effects can contribute to both components. The analytic fitted-feature moduli in the article supply an additional primitive account and are not misreported as a separately executed six-factor numerical experiment.

The R57 article adds the fitted-witness transfer theorem, its action-sensitive neural moduli and the new prospective experiment to the same NBO manuscript. The full preceding article, supplement, response, broader applications and unfavorable comparisons remain in the preserved and complete development documents. The release checks the inherited label sets and scientific-source/result identities, compiles all seven active and companion documents and rebuilds them from a clean archive. Those checks establish reproducibility and internal consistency, not an external editorial decision.

7. LOSSLESS NEURAL-SEARCH RECORDS

The new main-article results concern exact action recovery and the invariance of a mathematical prospective controller. This supplement identifies the finite computations used to evaluate them. The frozen R58 scientific code and protocol are preserved without modification; R59 adds the proof, manuscript integration and complete record-reconstruction audit. The audit does not retrain models or replace their measured service clocks.

7.1. *Feature reduction, arithmetic and fallback*

The stored ReLU coefficients and acquired-cell centers are interpreted as exact binary rationals. Action-dependent projections and activation-region tests are evaluated rationally. A ridge wholly inside one region on the full robust feasible interval is incorporated into the polynomial part, with its constant retained separately. Crossing ridges, including those with negative output weights, are evaluated by outward interval arithmetic. The expectation formula for a zero innovation projection is the ordinary positive part; division by a zero radius is not attempted.

The endpoint calculation evaluates every lattice index with outward conversion, arithmetic and pairwise summation. It does not use an unrounded matrix product as a rigorous endpoint. A point survives when its lower endpoint is no larger than the smallest upper endpoint. Exact rational evaluation of the original fitted objective, rather than its reduced floating-point midpoint, chooses the return and resolves ties. The implementation checks containment of the returned exact value in its interval.

The screening limit is 2049 lattice points. A larger lattice, nonfinite arithmetic or an invalid interval triggers the inherited exact algebraic solver. This fallback preserves exactness but can cost more than a successful screen; its occurrence is explicitly counted. The present production catalogue has no fallback. Small disjoint correctness fixtures exercise both fallback paths, exact ties, activation boundaries, signed weights, zero capacity, zero innovation projection, non-dyadic action spacing and exhaustive lattice agreement. These tests are not additional economic observations.

TABLE X: Exact activation-region reduction of trained ridges

(d, T)	Reduction	Crossing ridges	Reduced ridges	Survivors	Fallbacks
(2,2)	10%	37	987	32	0
(2,2)	20%	136	2936	96	0
(4,4)	10%	178	2894	96	0
(4,4)	20%	599	8617	288	0
(8,6)	10%	317	4803	160	0
(8,6)	20%	1054	14306	480	0

Counts aggregate ridge occurrences across neural queries and attempted resolutions, not the number of distinct trained features. Reduced ridges are incorporated exactly into polynomial coefficients or vanish identically. Their weights are not thresholded.

The survivor-gap and quadratic-growth statements are conditional analytical results. The production audit records actual survivors and endpoint hashes. It does not infer a global growth constant from those observed survivor counts. Similarly, the interval arithmetic count is distinct from a bit-complexity bound: exact rational operation cost depends on coefficient sizes, which are not assumed to be one machine word.

8. EVERY TIMING REPETITION AND PAID WORK COMPONENT

TABLE XI: Every complete-process timing repetition

(d, T)	Reduction	Mode	Rep. 1 (s)	Rep. 2 (s)	Rep. 3 (s)	Median (s)
(2,2)	10%	A	1.234	1.212	1.173	1.212
(2,2)	10%	S	1.097	1.101	1.147	1.101
(2,2)	20%	A	3.577	3.775	3.565	3.577
(2,2)	20%	S	3.374	3.409	3.640	3.409
(4,4)	10%	A	5.973	5.837	5.808	5.837
(4,4)	10%	S	5.487	5.494	5.456	5.487
(4,4)	20%	A	20.646	20.554	20.489	20.554
(4,4)	20%	S	19.373	19.396	19.344	19.373
(8,6)	10%	A	20.239	20.229	20.213	20.229
(8,6)	10%	S	19.785	19.223	19.270	19.270
(8,6)	20%	A	63.827	64.581	63.814	63.827
(8,6)	20%	S	61.842	62.079	62.075	62.075

No timing trial is selected or removed. A is algebraic and S screened. Fits and inference streams deliberately repeat; these are timing repetitions, not independent learned objects or additional policy-cost samples.

TABLE XII: Recorded service components and surrounding-process remainder

(d, T)	Reduction	Mode	Fit (s)	Search (s)	Verify (s)	Inference (s)	Other (s)
(2,2)	10%	A	0.122	0.178	0.367	0.146	0.384
(2,2)	10%	S	0.121	0.127	0.366	0.146	0.341
(2,2)	20%	A	0.466	0.495	1.043	1.188	0.381
(2,2)	20%	S	0.466	0.321	1.042	1.216	0.367
(4,4)	10%	A	0.492	0.878	3.907	0.127	0.435
(4,4)	10%	S	0.493	0.507	3.919	0.127	0.431
(4,4)	20%	A	1.891	2.530	11.607	3.999	0.492
(4,4)	20%	S	1.890	1.365	11.631	4.014	0.488
(8,6)	10%	A	1.057	1.626	15.742	1.247	0.536
(8,6)	10%	S	1.046	0.989	15.490	1.252	0.492
(8,6)	20%	A	4.398	4.901	47.688	6.243	0.638
(8,6)	20%	S	4.463	2.760	47.925	6.255	0.611

Entries are componentwise medians across the three services. Their sum need not equal the median complete-process clock. Other is calculated separately within each service as its surrounding clock minus the four component clocks; it includes imports, record handling, output and termination, not parent-controller checkpoint overhead. Search includes the common terminal solver.

TABLE XIII: Complete 36-service return, work and storage registry

(d, T)	Reduction	Mode	Rep.	Return	Attempts	Time (s)	MiB out	MiB RSS
(2,2)	20%	A	1	M	2	3.577	5.46	127.42
(2,2)	20%	A	2	M	2	3.775	5.46	127.77
(2,2)	20%	A	3	M	2	3.565	5.46	128.40
(2,2)	10%	A	1	M	1	1.234	0.73	80.54
(2,2)	10%	A	2	M	1	1.212	0.73	80.50
(2,2)	10%	A	3	M	1	1.173	0.73	80.50
(4,4)	20%	A	1	M	2	20.646	6.17	130.63
(4,4)	20%	A	2	M	2	20.554	6.17	130.07
(4,4)	20%	A	3	M	2	20.489	6.17	130.85
(4,4)	10%	A	1	M	1	5.973	0.43	80.56
(4,4)	10%	A	2	M	1	5.837	0.43	80.50
(4,4)	10%	A	3	M	1	5.808	0.43	80.69

	(d, T)	Reduction	Mode	Rep.	Return	Attempts	Time (s)	MiB out	MiB RSS	
1										1
2	(8,6)	20%	A	1	E	2	63.827	5.82	132.68	2
3	(8,6)	20%	A	2	E	2	64.581	5.82	133.25	3
4	(8,6)	20%	A	3	E	2	63.814	5.82	132.90	4
5	(8,6)	10%	A	1	M	1	20.239	1.26	81.27	5
6	(8,6)	10%	A	2	M	1	20.229	1.26	81.43	6
7	(8,6)	10%	A	3	M	1	20.213	1.26	81.48	7
8	(2,2)	20%	S	1	M	2	3.374	5.49	128.04	8
9	(2,2)	20%	S	2	M	2	3.409	5.49	127.97	9
10	(2,2)	20%	S	3	M	2	3.640	5.49	128.73	10
11	(2,2)	10%	S	1	M	1	1.097	0.74	80.70	11
12	(2,2)	10%	S	2	M	1	1.101	0.74	80.56	12
13	(2,2)	10%	S	3	M	1	1.147	0.74	80.59	13
14	(4,4)	20%	S	1	M	2	19.373	6.27	128.40	14
15	(4,4)	20%	S	2	M	2	19.396	6.27	128.80	15
16	(4,4)	20%	S	3	M	2	19.344	6.27	129.39	16
17	(4,4)	10%	S	1	M	1	5.487	0.46	80.42	17
18	(4,4)	10%	S	2	M	1	5.494	0.46	80.48	18
19	(4,4)	10%	S	3	M	1	5.456	0.46	80.62	19
20	(8,6)	20%	S	1	E	2	61.842	6.00	130.35	20
21	(8,6)	20%	S	2	E	2	62.079	6.00	130.79	21
22	(8,6)	20%	S	3	E	2	62.075	6.00	130.88	22
23	(8,6)	10%	S	1	M	1	19.785	1.32	81.41	23
24	(8,6)	10%	S	2	M	1	19.223	1.32	81.63	24
25	(8,6)	10%	S	3	M	1	19.270	1.32	81.32	25

M means target attained and E budget exhausted. Output bytes are complete serialized own-service files; RSS is process peak resident memory. These are not the logical working storage of the screening kernel. Exact service keys, parameter and policy identities, stagewise endpoints, raw arrays and clocks remain in the machine-readable audit. For each task, mode order alternates across repetition and target, processes run sequentially with fixed available-core affinity, and numerical-library thread counts are one. The three repetitions have the same fitted parameters and interval-path streams; they measure repeated work rather than independent training or inference variation. Governor availability and environment versions are

recorded in each service JSON. Frequency is not controlled, and different tasks use different runners.

The surrounding clock ends after process termination and log flush. It contains source-freeze verification, imports, fitting, all candidate search, verification, interval-path evaluation, inference and serialized service records. The controller then writes its checkpoint containing that elapsed value. This checkpoint operation is outside the surrounding service clock. Accordingly the table is complete for the measured service process and log, not a claim that the surrounding orchestration has no cost.

The component medians are descriptive. They need not sum to the median whole-process time. Output bytes refer to all files of the individual service, including intermediate attempts and raw arrays; process peak resident memory includes interpreter and numerical-library state. Neither replaces the vectorized screen’s logical working-storage bound.

9. DATEWISE ACCURACY AND THE INVARIANT STOPPING HISTORY

TABLE XIV: Datewise decisions and unchanged full-action certificate components

(d, T)	Reduction	Stage	Date	Changed	Extra	$U - L$	$U - C$	$C - L$
(2,2)	20%	0	0	30	0	0.40446	0.31866	0.08581
(2,2)	20%	0	1	27	0	0.65965	0.60626	0.05340
(2,2)	20%	1	0	62	1	0.31615	0.26553	0.05781
(2,2)	20%	1	1	58	0	0.49577	0.45576	0.04002
(2,2)	10%	0	0	30	0	0.40446	0.31866	0.08581
(2,2)	10%	0	1	27	0	0.65965	0.60626	0.05340
(4,4)	20%	0	0	31	0	0.74845	0.50284	0.24561
(4,4)	20%	0	1	31	0	0.69015	0.46336	0.22679
(4,4)	20%	0	2	28	0	0.59568	0.39956	0.19612
(4,4)	20%	0	3	15	0	0.69033	0.63704	0.05329
(4,4)	20%	1	0	63	0	0.68644	0.47865	0.20779
(4,4)	20%	1	1	63	0	0.63144	0.43964	0.19181
(4,4)	20%	1	2	59	0	0.54273	0.37687	0.16586
(4,4)	20%	1	3	42	0	0.68386	0.63602	0.04785
(4,4)	10%	0	0	31	0	0.74845	0.50284	0.24561

	(d, T)	Reduction	Stage	Date	Changed	Extra	$U - L$	$U - C$	$C - L$	
1										1
2	(4,4)	10%	0	1	31	0	0.69015	0.46336	0.22679	2
3	(4,4)	10%	0	2	28	0	0.59568	0.39956	0.19612	3
4	(4,4)	10%	0	3	15	0	0.69033	0.63704	0.05329	4
5	(8,6)	20%	0	0	26	0	1.02070	0.60302	0.41769	5
6	(8,6)	20%	0	1	24	0	0.99334	0.58712	0.40623	6
7	(8,6)	20%	0	2	24	0	0.94934	0.56153	0.38781	7
8	(8,6)	20%	0	3	20	0	0.87809	0.52013	0.35796	8
9	(8,6)	20%	0	4	14	0	0.76175	0.45268	0.30907	9
10	(8,6)	20%	0	5	0	0	0.71264	0.63956	0.07309	10
11	(8,6)	20%	1	0	60	0	0.95973	0.58520	0.37453	11
12	(8,6)	20%	1	1	60	0	0.93371	0.56954	0.36417	12
13	(8,6)	20%	1	2	59	0	0.89205	0.54452	0.34754	13
14	(8,6)	20%	1	3	54	0	0.82475	0.50409	0.32066	14
15	(8,6)	20%	1	4	46	0	0.71525	0.43849	0.27676	15
16	(8,6)	20%	1	5	0	0	0.70663	0.63853	0.06811	16
17	(8,6)	10%	0	0	26	0	1.02070	0.60302	0.41769	17
18	(8,6)	10%	0	1	24	0	0.99334	0.58712	0.40623	18
19	(8,6)	10%	0	2	24	0	0.94934	0.56153	0.38781	19
20	(8,6)	10%	0	3	20	0	0.87809	0.52013	0.35796	20
21	(8,6)	10%	0	4	14	0	0.76175	0.45268	0.30907	21
22	(8,6)	10%	0	5	0	0	0.71264	0.63956	0.07309	22

One of the exactly matching solver/repetition transcripts is printed to avoid duplicating numerical identities. Extra means a selected action changed by adding the exact fitted witness to the common plus critic-menu candidates. Each bound is rounded upward. The component maxima need not sum to the maximum total gap, and the local gap is not substituted for final Bellman loss. Every record retains the common candidate menu, the learned menu proposal and exact additional witness, together with the whole-cell candidate endpoints and continuous-action lower covers. The raw fields also retain the selected upper bound U , full-action lower bound L , intermediary C , selected action and the policy obtained without the additional exact witness. The identity

$$U - L = (U - C) + (C - L)$$

holds cellwise; maxima taken separately need not add to the maximum total. The extra-witness column identifies an actual change in the selected action, not merely a smaller fitted objective at the center.

These datewise quantities are local greedy-error certificates. The returned policy's all-state loss account also includes a valid bound for the installed reference continuation and is capped by the original total-cost support. The audit reconstructs that exact rational expression from every saved stage. Exact fitted optimization does not remove the full-action, acquisition, continuation or verification terms. The initial-law cost target and original Bellman-loss objective remain separate contracts.

For every saved inference look, the audit regenerates the declared bins, evaluates both the returned policy and installed zero policy on the corresponding continuous-bin intervals, and compares every lower and upper path endpoint. It verifies cost, gain and target-contrast bounds using the inherited exact-rational moment checks. The target classification is reconstructed from the exact endpoint, not from a rounded printed interval. A successful or positive-lower look forbids another look at that stage; success also forbids another construction attempt. Final exhaustion requires all declared attempts to have been used.

The reconstructed R58 catalogue has nine distinct task–target–attempt path streams. It has six distinct fitted-model verification inputs; both exact solvers are evaluated on each. All repeated instances still undergo file, parameter, policy, certificate and path-archive checks. Memoization avoids calling identical deterministic calculations independent observations. It does not omit an adverse service or permit an unmatched input to reuse another input's proof.

10. SOURCE PROVENANCE AND THE SCOPE OF THE REVISION

The controlling review is the 9 October 2026 advisory report on R54. The complete R57 manuscript supplies the original constructive operator, trained-neural transfer theory, non-tensor prospective method and conventional comparisons. The later R58 source-frozen execution supplies the matched exact-search experiment. R59 starts from the pinned R58 development commit and retains its entire scientific source and evidence history in the new revision directory.

The paper, supplement and response are ordinary sources; reproduction does not assemble mathematical claims from expiring downloads. The release checks inherited source/result hashes and labeled results, runs the previous test and audit chain, reconstructs the new matched catalogue, compiles the seven-document family and rebuilds it from a clean archive after removing generated outputs. The release manifest identifies the tested source and final remote package. These checks establish traceable execution and publication consistency, not external mathematical or editorial acceptance.

The economic observations retain their original cohort labels and confidence families. R59 publication adds zero training services and zero independent policy-cost samples. The equal-policy identity is exact, whereas the separate neural-versus-conventional comparisons retain their observed interval classifications. The broader NBO applications remain available under their own hypotheses and are not silently assigned the expected-cost theorem's assumptions.

11. EXACT SEARCH AND COMPLETE-SERVICE RECORDS

The new main-article statements are proved in the article. This supplement specifies their numerical objects and the full retained comparison. The original R59 main article, supplement, complete companions, response and source files remain available. The subsequent R60 scientific execution and its source freezes are immutable inputs, not results inferred from manuscript prose. R61 adds independent reconstruction, fixed-actor Bellman revalidation and a fresh language- and representation-matched timing experiment.

11.1. *Source and execution identities*

The controlling advisory review is the 9 October 2026 report on R59 manuscript commit 00412e6a4f43100996b324ea1ace097232c3fbbf. The complete corrected R60 scientific records are pinned to c55d18af14ee1573a8e356021d2326fae703cd23; the separately frozen centered-Bellman calculation is pinned to e94a8b867d6136ce14ba9697fc2288e0c7a7bc48. The R60 proof drafts are preserved separately from the integrated R61 proof text. Their consistency clause is made explicit about

fixed-cover endpoint stability; the new lemma establishes that condition for the implementation rather than assuming that outward inclusion alone supplies it.

The primary reconstruction checks 5,600 protected predecessor files and 2,141 scientific input files by their recorded identities. It reconstructs 2,408 scalar solver records, 32 two-control query records, 144 service stages and 24,576 cell-date decisions. It checks 352 stopping looks and 1,128 confidence-interval records. Identical numerical inputs are cached by their complete parameters, acquired geometry and declared path stream. The resulting 1,896,448 reconstructed path rows are a verification count, not new independent observations or an augmentation of the original statistical sample size.

The new native-factorial audit checks 3160 exact answers against independent Fraction exhaustive evaluation. These include repeated queries and answers returned in different batches. Its source freeze precedes the fresh timing measurements; its instance set was specified after the R60 outcomes had been observed, and this information set is recorded in the protocol. A successful exact check does not turn a hosted timing into a hardware-independent complexity theorem.

11.2. *Every prospective return*

TABLE XV: All ninety-six original R60 service returns

Host	d/T	q	Seed	Mode	Stages	Met	Work (s)	Cost upper	Old gap
0	2/2	9/10	60103	C	1	Y	1.253	0.64900	3.94572
0	2/2	9/10	60103	Q	1	Y	1.308	0.64900	3.94572
0	2/2	9/10	60103	N	1	Y	1.527	0.64900	3.94572
0	2/2	9/10	60103	S	1	Y	1.563	0.64900	3.94572
0	2/2	9/10	60209	C	1	Y	1.236	0.64850	3.94572
0	2/2	9/10	60209	Q	1	Y	1.312	0.64850	3.94572
0	2/2	9/10	60209	N	1	Y	1.537	0.64851	3.94572
0	2/2	9/10	60209	S	1	Y	1.561	0.64851	3.94572
0	2/2	4/5	60103	C	2	Y	3.781	0.62178	3.85741
0	2/2	4/5	60103	Q	2	Y	3.979	0.62178	3.85741
0	2/2	4/5	60103	N	2	Y	4.773	0.62177	3.85741

	Host	d/T	q	Seed	Mode	Stages	Met	Work (s)	Cost upper	Old gap	
1											1
2	0	2/2	4/5	60103	S	2	Y	4.805	0.62177	3.85741	2
3	0	2/2	4/5	60209	C	2	Y	3.802	0.61871	3.85741	3
4	0	2/2	4/5	60209	Q	2	Y	3.983	0.61871	3.85741	4
5	0	2/2	4/5	60209	N	2	Y	4.790	0.61843	3.85741	5
6	0	2/2	4/5	60209	S	2	Y	4.835	0.61843	3.85741	6
7	0	4/4	9/10	60103	C	1	Y	4.371	1.06500	6.78416	7
8	0	4/4	9/10	60103	Q	1	Y	4.704	1.06500	6.78416	8
9	0	4/4	9/10	60103	N	1	Y	5.436	1.06486	6.78416	9
10	0	4/4	9/10	60103	S	1	Y	5.528	1.06486	6.78416	10
11	0	4/4	9/10	60209	C	1	Y	4.323	1.07342	6.78416	11
12	0	4/4	9/10	60209	Q	1	Y	4.773	1.07342	6.78416	12
13	0	4/4	9/10	60209	N	1	Y	5.473	1.07342	6.78416	13
14	0	4/4	9/10	60209	S	1	Y	5.518	1.07342	6.78416	14
15	0	4/4	4/5	60103	C	2	Y	15.777	0.97214	6.72215	15
16	0	4/4	4/5	60103	Q	2	Y	17.049	0.97214	6.72215	16
17	0	4/4	4/5	60103	N	2	Y	19.527	0.97210	6.72215	17
18	0	4/4	4/5	60103	S	2	Y	19.734	0.97210	6.72215	18
19	0	4/4	4/5	60209	C	2	Y	15.893	0.97177	6.72215	19
20	0	4/4	4/5	60209	Q	2	Y	16.974	0.97177	6.72215	20
21	0	4/4	4/5	60209	N	2	Y	19.521	0.97177	6.72215	21
22	0	4/4	4/5	60209	S	2	Y	19.859	0.97177	6.72215	22
23	0	8/6	9/10	60103	C	1	Y	16.851	1.47572	9.24880	23
24	0	8/6	9/10	60103	Q	1	Y	18.316	1.47572	9.24880	24
25	0	8/6	9/10	60103	N	1	Y	19.664	1.47572	9.24880	25
26	0	8/6	9/10	60103	S	1	Y	19.592	1.47572	9.24880	26
27	0	8/6	9/10	60209	C	1	Y	16.866	1.47436	9.24880	27
28	0	8/6	9/10	60209	Q	1	Y	18.427	1.47436	9.24880	28
29	0	8/6	9/10	60209	N	1	Y	19.486	1.47436	9.24880	29
30	0	8/6	9/10	60209	S	1	Y	19.759	1.47436	9.24880	30
31	0	8/6	4/5	60103	C	2	N	52.752	1.36361	9.18783	31
32	0	8/6	4/5	60103	Q	2	N	57.263	1.36361	9.18783	32
	0	8/6	4/5	60103	N	2	N	61.488	1.36361	9.18783	
	0	8/6	4/5	60103	S	2	N	62.227	1.36361	9.18783	

	Host	d/T	q	Seed	Mode	Stages	Met	Work (s)	Cost upper	Old gap	
1											1
2	0	8/6	4/5	60209	C	2	N	52.649	1.36272	9.18783	2
3	0	8/6	4/5	60209	Q	2	N	57.301	1.36272	9.18783	3
4	0	8/6	4/5	60209	N	2	N	62.051	1.36272	9.18783	4
5	0	8/6	4/5	60209	S	2	N	62.660	1.36272	9.18783	5
6	1	2/2	9/10	60103	C	1	Y	1.249	0.64900	3.94572	6
7	1	2/2	9/10	60103	Q	1	Y	1.308	0.64900	3.94572	7
8	1	2/2	9/10	60103	N	1	Y	1.527	0.64900	3.94572	8
9	1	2/2	9/10	60103	S	1	Y	1.579	0.64900	3.94572	9
10	1	2/2	9/10	60209	C	1	Y	1.245	0.64850	3.94572	10
11	1	2/2	9/10	60209	Q	1	Y	1.312	0.64850	3.94572	11
12	1	2/2	9/10	60209	N	1	Y	1.547	0.64851	3.94572	12
13	1	2/2	9/10	60209	S	1	Y	1.568	0.64851	3.94572	13
14	1	2/2	4/5	60103	C	2	Y	3.792	0.62178	3.85741	14
15	1	2/2	4/5	60103	Q	2	Y	3.960	0.62178	3.85741	15
16	1	2/2	4/5	60103	N	2	Y	4.721	0.62177	3.85741	16
17	1	2/2	4/5	60103	S	2	Y	4.840	0.62177	3.85741	17
18	1	2/2	4/5	60209	C	2	Y	3.794	0.61871	3.85741	18
19	1	2/2	4/5	60209	Q	2	Y	3.951	0.61871	3.85741	19
20	1	2/2	4/5	60209	N	2	Y	4.738	0.61843	3.85741	20
21	1	2/2	4/5	60209	S	2	Y	4.796	0.61843	3.85741	21
22	1	4/4	9/10	60103	C	1	Y	4.395	1.06500	6.78416	22
23	1	4/4	9/10	60103	Q	1	Y	4.716	1.06500	6.78416	23
24	1	4/4	9/10	60103	N	1	Y	5.475	1.06486	6.78416	24
25	1	4/4	9/10	60103	S	1	Y	5.588	1.06486	6.78416	25
26	1	4/4	9/10	60209	C	1	Y	4.337	1.07342	6.78416	26
27	1	4/4	9/10	60209	Q	1	Y	4.708	1.07342	6.78416	27
28	1	4/4	9/10	60209	N	1	Y	5.453	1.07342	6.78416	28
29	1	4/4	9/10	60209	S	1	Y	5.527	1.07342	6.78416	29
30	1	4/4	4/5	60103	C	2	Y	15.923	0.97214	6.72215	30
31	1	4/4	4/5	60103	Q	2	Y	16.783	0.97214	6.72215	31
32	1	4/4	4/5	60103	N	2	Y	19.374	0.97210	6.72215	32
	1	4/4	4/5	60103	S	2	Y	19.671	0.97210	6.72215	
	1	4/4	4/5	60209	C	2	Y	15.852	0.97177	6.72215	

1	Host	d/T	q	Seed	Mode	Stages	Met	Work (s)	Cost upper	Old gap	1
2	1	4/4	4/5	60209	Q	2	Y	17.025	0.97177	6.72215	2
3	1	4/4	4/5	60209	N	2	Y	20.341	0.97177	6.72215	3
4	1	4/4	4/5	60209	S	2	Y	19.746	0.97177	6.72215	4
5	1	8/6	9/10	60103	C	1	Y	16.772	1.47572	9.24880	5
6	1	8/6	9/10	60103	Q	1	Y	18.201	1.47572	9.24880	6
7	1	8/6	9/10	60103	N	1	Y	19.527	1.47572	9.24880	7
8	1	8/6	9/10	60103	S	1	Y	19.619	1.47572	9.24880	8
9	1	8/6	9/10	60209	C	1	Y	16.802	1.47436	9.24880	9
10	1	8/6	9/10	60209	Q	1	Y	18.341	1.47436	9.24880	10
11	1	8/6	9/10	60209	N	1	Y	19.204	1.47436	9.24880	11
12	1	8/6	9/10	60209	S	1	Y	19.642	1.47436	9.24880	12
13	1	8/6	4/5	60103	C	2	N	52.077	1.36361	9.18783	13
14	1	8/6	4/5	60103	Q	2	N	57.191	1.36361	9.18783	14
15	1	8/6	4/5	60103	N	2	N	61.391	1.36361	9.18783	15
16	1	8/6	4/5	60103	S	2	N	62.815	1.36361	9.18783	16
17	1	8/6	4/5	60209	C	2	N	52.187	1.36272	9.18783	17
18	1	8/6	4/5	60209	Q	2	N	57.769	1.36272	9.18783	18
19	1	8/6	4/5	60209	N	2	N	61.548	1.36272	9.18783	19
20	1	8/6	4/5	60209	S	2	N	62.456	1.36272	9.18783	20

No service is omitted. Cost upper is the final original-law simultaneous confidence endpoint; Old gap is the maximum datewise original certificate before the new fixed-policy revalidation. The two are different estimands. Modes and complete-work semantics are those of the main service table. A failed target still incurs all executed work. Raw lower endpoints, all intermediate looks, nonterminal changes and stage costs remain in the ordinary result registry.

Each service retains its full fitted-parameter payload, training diagnostics, acquired partition, candidate indices, exact action witness, policy identity, whole-cell lower and upper advantage arrays, complete continuous-action covers and all sequential inference endpoints. The parent recorder retains the randomized within-design method order, child process log, first durable completion receipt and

immutable numbered checkpoints. A byte-count check includes all recorded child files. The final ledger is not substituted for the first-return timing boundary.

The first R60 attempt failed when its exclusive-create parent writer revisited the same ledger path. No mathematical child-service change was needed for the corrected recorder. Its partial results and failure logs remain under the original attempt directory. The complete corrected services are reruns with a separately frozen parent recorder, not selected successful fragments of the first attempt. Study-level recovery and reproduction overhead are retained separately from the reported complete child-return cost. The new publication adds no training service or economic path observation by rerunning these records.

11.3. *Inference and policy identity*

Conditional on the stored fitted objects and the declared selection information, the original path calculation uses outward enclosures over the original continuous initial and innovation laws. The fixed seed identifies a reproducible pseudorandom stream; the statistical independence model is not a mathematical consequence of the seed. Common paths induce dependence between policies, which is retained in paired contrasts. Negative pathwise gains are not clipped on the ground that a statewise conditional-expectation improvement was proved.

Stopping follows the frozen sequence of sample looks and construction rungs. Every intermediate failure and every final budget exhaustion is checked. Exact native and screened witnesses give the same prospective transcript because the controller uses their mathematical outputs, not a solver-specific elapsed-time cutoff. Replacing the solver in a controller that instead stops at a wall-clock limit would require a different transcript argument. The fresh final-policy comparison has its own retained stream and coverage account; its inference work is additional study overhead, not an unobserved marginal cost assigned to one earlier service.

12. COMPLETE IMPLEMENTATION ABLATIONS

The four native cells share the same arbitrary-size rational engine, exact original objective and smallest-index tie rule. Reduction changes only an algebraically equivalent representation plus a recovered constant. The exact final original-objective

TABLE XVI

MATCHED NATIVE TWO-BY-TWO REPETITION MEDIAN

Host	Regime	UE (ms)	RE (ms)	UP (ms)	RP (ms)
0	mixed	4.528	3.327	3.269	2.782
0	crossing	4.484	4.697	4.036	4.202
0	cancellation	4.280	4.431	2.960	3.069
0	near-tie	4.444	4.598	3.020	3.175
0	flat	4.289	4.430	2.980	3.142
1	mixed	4.400	3.105	3.132	2.556
1	crossing	4.464	4.486	3.873	4.014
1	cancellation	4.235	4.326	2.789	2.885
1	near-tie	4.358	4.529	2.848	2.965
1	flat	4.151	4.290	2.741	2.882

U/R indicates unreduced/reduced activation representation; E/P indicates exhaustive/finite-difference native search. All four use the same exact engine, lattice, tie rule and timing boundary. Seven additional shuffled repetitions use seed 60103, width 32 and lattice 129 in every regime. Full min/max ranges and per-repetition orders are in FACTORIAL_AUDIT61.json and the two source summaries. No repeated instance is counted as a new learned policy.

TABLE XVII

WORK FOR THE SAME NINETY-QUERY CATALOGUE AT DIFFERENT NATIVE BATCH SIZES

Host	Batch	Processes	UE (s)	RE (s)	UP (s)	RP (s)
0	1	90	0.4339	0.4084	0.2216	0.2243
0	8	12	0.3134	0.2993	0.1152	0.1173
0	32	3	0.3006	0.2888	0.1020	0.1042
1	1	90	0.4175	0.3998	0.2085	0.2093
1	8	12	0.3233	0.3071	0.1161	0.1146
1	32	3	0.3100	0.2957	0.1013	0.1030

Each entry covers all ninety heterogeneous fixed queries, including exact reduction when applicable, serialization, process creation, parsing and original-objective checks. It is not a per-query median or a complete policy-service clock. The entire answer arrays coincide with Fraction exhaustive references for every batch size. Compilation is separately recorded.

check is charged in every cell, including exhaustive search. The reduced exhaustive versus reduced piecewise contrast isolates the search change at a common reduced representation. The unreduced pair and the reduction contrasts expose the remaining mechanisms. No speed difference between Python symbolic work and native arithmetic is described as a pure interval-screening effect.

The complete batch input contains ninety heterogeneous fixed rational queries. Every query is retained in the same declared order within a batch-size calculation; the method order is randomized within each batch-size block. Batch size one creates ninety processes, size eight creates twelve, and size thirty-two creates three. Input reduction, encoding, process calls, output decoding and original-value checks are measured. Native process time and request and response byte counts are also retained. The ordinary per-service code has its own batching, and the batch catalogue does not retime it retrospectively.

TABLE XVIII: Complete eight-method stress summary

Host	Regime	Method	Grid ms	Repeat ms	Fallback	Survive	Bits	Bytes
0	mixed	SY	20.706	26.682	0	0	44	–
0	mixed	RS	12.655	21.612	0	0	44	–
0	mixed	VE	6.276	7.257	0	1	44	155952
0	mixed	SU	4.267	4.937	0	1	44	262656
0	mixed	SR	4.059	4.468	0	1	44	155952
0	mixed	UE	4.137	5.059	0	0	44	–
0	mixed	RP	2.737	3.569	0	0	44	–
0	mixed	AD	6.411	20.120	1	0	44	–
0	crossing	SY	28.554	40.938	0	0	44	–
0	crossing	RS	23.186	40.724	0	0	44	–
0	crossing	VE	6.504	7.710	0	1	44	262656
0	crossing	SU	4.550	5.092	0	1	44	262656
0	crossing	SR	4.315	4.789	0	1	44	262656
0	crossing	UE	4.205	5.226	0	0	44	–
0	crossing	RP	3.758	4.597	0	0	44	–
0	crossing	AD	11.385	40.056	1	0	44	–

	Host	Regime	Method	Grid ms	Repeat ms	Fallback	Survive	Bits	Bytes		
1		0	cancellation	SY	11.890	20.303	0	0	44	–	1
2		0	cancellation	RS	12.339	20.430	0	0	44	–	2
3		0	cancellation	VE	6.340	7.807	0	1	57	262656	3
4		0	cancellation	SU	4.215	4.855	0	1	57	262656	4
5		0	cancellation	SR	4.324	5.040	0	1	57	262656	5
6		0	cancellation	UE	3.901	4.750	0	0	57	–	6
7		0	cancellation	RP	3.064	3.682	0	0	57	–	7
8		0	cancellation	AD	32.294	103.697	11	0	57	–	8
9		0	near-tie	SY	12.798	21.347	0	0	42	–	9
10		0	near-tie	RS	13.122	21.298	0	0	42	–	10
11		0	near-tie	VE	6.255	8.068	0	2	42	262656	11
12		0	near-tie	SU	4.167	4.941	0	2	35	262656	12
13		0	near-tie	SR	4.567	5.009	0	2	35	262656	13
14		0	near-tie	UE	4.006	4.946	0	0	42	–	14
15		0	near-tie	RP	2.951	3.947	0	0	42	–	15
16		0	near-tie	AD	24.964	59.917	4	0	42	–	16
17		0	flat	SY	6.361	10.717	0	0	1	–	17
18		0	flat	RS	5.997	10.883	0	0	1	–	18
19		0	flat	VE	6.231	7.747	0	513	24	262656	19
20		0	flat	SU	6.214	7.434	0	513	24	262656	20
21		0	flat	SR	6.148	7.714	0	513	24	262656	21
22		0	flat	UE	3.822	5.099	0	0	24	–	22
23		0	flat	RP	2.831	3.642	0	0	24	–	23
24		0	flat	AD	31.511	98.848	11	0	24	–	24
25		1	mixed	SY	14.148	26.007	0	0	44	–	25
26		1	mixed	RS	16.803	21.079	0	0	44	–	26
27		1	mixed	VE	5.466	6.786	0	1	44	155952	27
28		1	mixed	SU	3.839	4.325	0	1	44	262656	28
29		1	mixed	SR	3.743	3.846	0	1	44	155952	29
30		1	mixed	UE	3.492	4.867	0	0	44	–	30
31		1	mixed	RP	2.247	2.828	0	0	44	–	31
32		1	mixed	AD	6.529	19.985	1	0	44	–	32
		1	crossing	SY	23.279	40.619	0	0	44	–	

	Host	Regime	Method	Grid ms	Repeat ms	Fallback	Survive	Bits	Bytes	
1										1
2	1	crossing	RS	21.393	40.478	0	0	44	–	2
3	1	crossing	VE	6.055	7.631	0	1	44	262656	3
4	1	crossing	SU	3.874	4.668	0	1	44	262656	4
5	1	crossing	SR	4.176	4.404	0	1	44	262656	5
6	1	crossing	UE	4.328	4.988	0	0	44	–	6
7	1	crossing	RP	3.123	4.717	0	0	44	–	7
8	1	crossing	AD	11.500	39.973	1	0	44	–	8
9	1	cancellation	SY	11.982	20.002	0	0	44	–	9
10	1	cancellation	RS	12.148	19.945	0	0	44	–	10
11	1	cancellation	VE	5.654	7.018	0	1	57	262656	11
12	1	cancellation	SU	3.766	4.448	0	1	57	262656	12
13	1	cancellation	SR	3.842	4.517	0	1	57	262656	13
14	1	cancellation	UE	3.543	4.446	0	0	57	–	14
15	1	cancellation	RP	2.594	3.206	0	0	57	–	15
16	1	cancellation	AD	31.958	102.212	11	0	57	–	16
17	1	near-tie	SY	12.740	21.076	0	0	42	–	17
18	1	near-tie	RS	12.228	21.059	0	0	42	–	18
19	1	near-tie	VE	5.818	7.278	0	2	42	262656	19
20	1	near-tie	SU	3.846	4.242	0	2	35	262656	20
21	1	near-tie	SR	3.819	4.478	0	2	35	262656	21
22	1	near-tie	UE	3.689	4.634	0	0	42	–	22
23	1	near-tie	RP	2.630	3.319	0	0	42	–	23
24	1	near-tie	AD	24.696	60.688	4	0	42	–	24
25	1	flat	SY	6.586	10.663	0	0	1	–	25
26	1	flat	RS	5.783	10.665	0	0	1	–	26
27	1	flat	VE	5.687	6.950	0	513	24	262656	27
28	1	flat	SU	5.763	6.839	0	513	24	262656	28
29	1	flat	SR	5.691	6.920	0	513	24	262656	29
30	1	flat	UE	3.581	4.315	0	0	24	–	30
31	1	flat	RP	2.666	3.160	0	0	24	–	31
32	1	flat	AD	31.565	98.507	11	0	24	–	32

SY and RS are unreduced and reduced symbolic searches; VE computes the vector enclosure but evaluates every lattice action exactly; SU and SR screen the unreduced and reduced objectives; UE is unreduced native exhaustive search; RP is reduced native finite-difference search; AD is reduced adaptive interval subdivision with an exact fallback. Grid ms is the median over all eighteen seed/width/lattice instances in that regime; Repeat ms is the fixed seven-repeat subset median. Survive, Bits and Bytes are maxima of the named diagnostics, not maxima over every hidden arithmetic temporary or whole-process memory. The observed endpoint intervention coarsens exact bounds to 8 or 24 fractional bits; it does not change internal binary64 precision.

TABLE XIX: Every large-lattice and node-budget guard case

Host	Lattice	Regime	Method	Fallback	Nodes	Exact total	Time (s)
0	2050	mixed	RP	N	0	20	0.0029
0	2050	mixed	SR	Y	0	20	0.0024
0	2050	mixed	AD	N	49	4	0.0054
0	2050	cancellation	RP	N	0	19	0.0027
0	2050	cancellation	SR	Y	0	19	0.0022
0	2050	cancellation	AD	Y	256	22	0.0346
0	2050	near-tie	RP	N	0	19	0.0027
0	2050	near-tie	SR	Y	0	19	0.0023
0	2050	near-tie	AD	Y	256	22	0.0331
0	2050	flat	RP	N	0	18	0.0028
0	2050	flat	SR	Y	0	18	0.0022
0	2050	flat	AD	Y	256	21	0.0337
0	4097	mixed	RP	N	0	15	0.0026
0	4097	mixed	SR	Y	0	15	0.0021
0	4097	mixed	AD	N	117	3	0.0111
0	4097	cancellation	RP	N	0	9	0.0026
0	4097	cancellation	SR	Y	0	9	0.0022
0	4097	cancellation	AD	Y	256	12	0.0310
0	4097	near-tie	RP	N	0	9	0.0028
0	4097	near-tie	SR	Y	0	9	0.0022
0	4097	near-tie	AD	Y	256	12	0.0340

	Host	Lattice	Regime	Method	Fallback	Nodes	Exact total	Time (s)	
1									1
2	0	4097	flat	RP	N	0	8	0.0026	2
3	0	4097	flat	SR	Y	0	8	0.0022	3
4	0	4097	flat	AD	Y	256	11	0.0295	4
5	1	2050	mixed	RP	N	0	20	0.0024	5
6	1	2050	mixed	SR	Y	0	20	0.0019	6
7	1	2050	mixed	AD	N	49	4	0.0057	7
8	1	2050	cancellation	RP	N	0	19	0.0026	8
9	1	2050	cancellation	SR	Y	0	19	0.0020	9
10	1	2050	cancellation	AD	Y	256	22	0.0351	10
11	1	2050	near-tie	RP	N	0	19	0.0025	11
12	1	2050	near-tie	SR	Y	0	19	0.0019	12
13	1	2050	near-tie	AD	Y	256	22	0.0328	13
14	1	2050	flat	RP	N	0	18	0.0025	14
15	1	2050	flat	SR	Y	0	18	0.0019	15
16	1	2050	flat	AD	Y	256	21	0.0329	16
17	1	4097	mixed	RP	N	0	15	0.0025	17
18	1	4097	mixed	SR	Y	0	15	0.0019	18
19	1	4097	mixed	AD	N	117	3	0.0112	19
20	1	4097	cancellation	RP	N	0	9	0.0024	20
21	1	4097	cancellation	SR	Y	0	9	0.0018	21
22	1	4097	cancellation	AD	Y	256	12	0.0315	22
23	1	4097	near-tie	RP	N	0	9	0.0024	23
24	1	4097	near-tie	SR	Y	0	9	0.0019	24
25	1	4097	near-tie	AD	Y	256	12	0.0331	25
26	1	4097	flat	RP	N	0	8	0.0024	26
27	1	4097	flat	SR	Y	0	8	0.0018	27
28	1	4097	flat	AD	Y	256	11	0.0289	28

30 The lattice sizes exceed the declared screening threshold. An exact fallback preserves the original value
 31 and smallest witness but incurs the already spent prefix work. Exact total adds the explicitly recorded
 32 pre-fallback evaluations. Native rational operation counts remain distinct from wall time.

The stress regimes have different mathematical meanings. Crossing features remain active on an action interval. Signed cancellation uses large opposing terms whose true objective difference can be small. The near-tie fixture creates an exact equality between adjacent lattice actions, while flat objectives have many exact minimizers. Neither is silently perturbed to select a unique minimizer. The coarse-endpoint diagnostics round valid exact bounds outward to a prescribed fractional-bit resolution. They test the effect of wider screening bounds, not the numerical semantics of an unexecuted lower-precision processor.

The stored maximum operand-bit diagnostic observes declared inputs, selected polynomial coefficients and selected rational evaluations. It is not a trace of every temporary value inside the arbitrary-size arithmetic library. The mathematical bit-cost statement therefore uses an explicit bound on the actual operand lengths and a separate rational-operation cost, while the data report the measured diagnostic under its own name. Action lattice indices in the executed C++ backend use a machine integer; all declared indices fit that range. Extending the numerical catalogue beyond that range requires a correspondingly adequate index representation, even though the rational coefficients already have arbitrary precision.

The sparse endpoint screen and the adaptive solver have different fallback rules. A screening guard can reject a requested full array before allocating it. Adaptive subdivision can spend its prefix node budget and then run an exact fallback. Both preserve the original objective and first witness. Work accounting includes the unsuccessful prefix in the latter case. The maximum size of a recorded interval array is not total process resident memory; the full service record separately reports parent and child peak resident memory.

13. MULTI-ACTION INTEGRATION AND CONSTRAINED RECOVERY

TABLE XX: All two-control exact query records

Host	Seed	State	Cap	Shocks	Witness	Exhaust.	Adaptive	Nodes	Fall.
0	60103	1/8,1/4	8	1	3,5	45	7	55	N
0	60103	1/8,1/4	8	2	3,5	45	7	55	N

	Host	Seed	State	Cap	Shocks	Witness	Exhaust.	Adaptive	Nodes	Fall.	
1											1
2	0	60103	1/8,1/4	16	1	6,10	153	7	113	N	2
3	0	60103	1/8,1/4	16	2	6,10	153	7	113	N	3
4	0	60103	3/4,5/8	8	1	1,7	45	7	105	N	4
5	0	60103	3/4,5/8	8	2	1,7	45	7	105	N	5
6	0	60103	3/4,5/8	16	1	1,14	153	153	256	Y	6
7	0	60103	3/4,5/8	16	2	1,14	153	153	256	Y	7
8	0	60209	1/8,1/4	8	1	2,6	45	6	55	N	8
9	0	60209	1/8,1/4	8	2	2,6	45	6	55	N	9
10	0	60209	1/8,1/4	16	1	5,11	153	7	111	N	10
11	0	60209	1/8,1/4	16	2	5,11	153	7	111	N	11
12	0	60209	3/4,5/8	8	1	1,2	45	8	107	N	12
13	0	60209	3/4,5/8	8	2	1,2	45	8	107	N	13
14	0	60209	3/4,5/8	16	1	1,3	153	153	256	Y	14
15	0	60209	3/4,5/8	16	2	1,3	153	153	256	Y	15
16	1	60103	1/8,1/4	8	1	3,5	45	7	55	N	16
17	1	60103	1/8,1/4	8	2	3,5	45	7	55	N	17
18	1	60103	1/8,1/4	16	1	6,10	153	7	113	N	18
19	1	60103	1/8,1/4	16	2	6,10	153	7	113	N	19
20	1	60103	3/4,5/8	8	1	1,7	45	7	105	N	20
21	1	60103	3/4,5/8	8	2	1,7	45	7	105	N	21
22	1	60103	3/4,5/8	16	1	1,14	153	153	256	Y	22
23	1	60103	3/4,5/8	16	2	1,14	153	153	256	Y	23
24	1	60209	1/8,1/4	8	1	2,6	45	6	55	N	24
25	1	60209	1/8,1/4	8	2	2,6	45	6	55	N	25
26	1	60209	1/8,1/4	16	1	5,11	153	7	111	N	26
27	1	60209	1/8,1/4	16	2	5,11	153	7	111	N	27
28	1	60209	3/4,5/8	8	1	1,2	45	8	107	N	28
29	1	60209	3/4,5/8	8	2	1,2	45	8	107	N	29
30	1	60209	3/4,5/8	16	1	1,3	153	153	256	Y	30
31	1	60209	3/4,5/8	16	2	1,3	153	153	256	Y	31

Each original rational objective, trained-parameter payload, full exhaustive reference, adaptive answer and clock is retained. The additional action and optional second shock are explicit nested model

extensions; the original scalar-action economy is recovered by zero second action with the second shock disabled.

For independent box-uniform components, repeated integration of the positive part gives the exact signed vertex formula in the main article. Zero radii are removed before division. A finite mixture is evaluated with nonnegative rational mixture weights. Signed network output weights reverse interval endpoints when necessary. This operation is not an assumption that the sum of two uniforms is itself uniform.

The two-control objective retains both nonlinear investment costs, the stated cross term, both exposure columns and the shared capacity constraint. The original scalar action-dependent objective is recovered on the stated nested restriction. Every feasible lattice pair is evaluated by the exhaustive reference. Subdivision starts from a feasible exact incumbent, prunes only with a valid lower bound and lexicographic tie test, and can fall back to the same exhaustive problem. The second-action and second-shock query records are not relabeled as full policy sweeps or as a calibrated economic experiment.

14. ORIGINAL-OPTIMUM BRACKETS AND FIXED-POLICY REVALIDATION

The common-reference construction keeps $J_t^* - J_t^0$ and $J_t^\pi - J_t^0$ on the same state. Its narrower bracket is not obtained by changing the initial law, replacing continuous feasible actions by a finite comparison menu, or dropping the cost of a future actor. Complete continuous-action interval covers supply the Bellman lower bound. Only robustly feasible quantum actions construct its upper policy. Statewise residual differences and original-law innovation bins provide the signed continuation calculation without a policy-value oracle.

The refinement grid (N, A, q) follows the sequence

$$(8, 8, 4), \quad (16, 16, 8), \quad (32, 32, 16), \\ (64, 64, 32), \quad (128, 64, 32).$$

The final action quantum stays fixed, and the last action-cover count is held at 64. The measured ladder therefore has a stated residual tolerance, not a claim of an executed arbitrary-accuracy limit. The consistency theorem separately requires

uniformly vanishing local approximation and arithmetic allowances and stable
endpoint maps, including refinement of a binding action quantum when needed.

The new fixed-policy calculation includes all thirty-two original two-date service
aliases. Reconstruction of the full policy-plus-partition identity yields five distinct
actors; equality of one expected cost is not used for deduplication. The verification
grid is fixed before these actor bounds are calculated. Every fine cell includes all
intersecting acquired leaves and both closed endpoints. No smoothness hypothesis
is imposed on a discontinuous actor, and no midpoint action is silently assigned to
a cell crossed by an actor boundary.

The nonpositive reference-advantage gate is independently reconstructed at
every original acquired cell and date before the zero-excess bound is used. The
upper recursion and its unclipped counterpart are then both recomputed from the
primitives, and both endpoint arrays are retained. Their maximum all-date gaps
happen to coincide in this catalogue; that is an observed result, not a reason to omit
the unclipped calculation. Original policies, seeds, partitions, stopping targets and
inference samples are unchanged.

Each actor's additional cost covers the complete upper recursion and endpoint
archive. A common lower-table reconstruction is shared verification work. The outer
process record includes imports, the log and its first durable receipt. These clocks
refer to their actual executions. The original centered ladder's construction time,
a later reconstruction time and an earlier prospective service on another worker
are not summed and called one observed end-to-end run. Such components can
be separately budgeted for a future deployment, but that is a proposed allocation
rather than a measured historical service.

15. PUBLICATION AND PRESERVATION

The publication script uses ordinary committed manuscript sources. It verifies
the predecessor preservation maps and all new source freezes, reruns the exact
theorem and implementation regressions, checks every inherited labeled result,
regenerates the current tables and compiles the seven-document package. A clean
archive rebuild removes generated PDFs and current tables before repeating the

ordinary build. Output page counts and extracted-text identities are compared, and every delivered source, evidence file and PDF has a final identity record.

New proof statements and the fixed-cover consistency clarification are distinguished from inherited results. The original R59 article and introduction files remain preserved, as do all original economic applications and adverse numerical comparisons in the complete development companions. The active introduction now supplies a single reading path rather than repeated revision prefaces. The title and economic research subject remain Neural Bellman Operators. Compilation, source identity, exact regression tests and deterministic reproduction are evidence about the delivered calculation, not an external mathematical acceptance decision.

REFERENCES